Naoki Masuda

Contact Information

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Employment

7/2025–	Professor (tenured) Gilbert S. Omenn Department of Computational Medicine and Informatics & Department of Mathematics, University of Michigan, USA
9/2021–6/2025	Professor (tenured) Department of Mathematics, State University of New York at Buffalo, USA Director of Graduate Studies, 7/2022–6/2025
8/19/2019–8/2021	Associate Professor (non-tenured) Department of Mathematics, State University of New York at Buffalo, USA
8/19/2019–	Institute for Artificial Intelligence and Data Science (incorporating the Computational and Data-Enabled Sciences and Engineering Program, which I have been core faculty of, in 4/2021) State University of New York at Buffalo, USA
8/1/2019– 8/18/2019	Associate Professor (tenured) Department of Engineering Mathematics, University of Bristol, United Kingdom
3/2014–7/2019	Senior Lecturer (tenured after one year of probation) Department of Engineering Mathematics, University of Bristol, United Kingdom
9/2008–2/2014	Associate Professor (tenured) Department of Mathematical Informatics, University of Tokyo, Japan
10/2006–8/2008	Lecturer (tenured) Department of Mathematical Informatics, University of Tokyo, Japan
4/2004–9/2006	Special Postdoctoral Fellow Amari Research Unit, RIKEN Brain Science Institute, Japan Host researcher: Professor Shun-ichi Amari
4/2003–3/2004	Research Fellow (PD) Japan Society for the Promotion of Science, at Yokohama National University, Japan Host researcher: Professor Norio Konno
10/2002–3/2003	Research Fellow (PD) Japan Society for the Promotion of Science, at University of Tokyo, Japan Host researcher: Professor Kazuyuki Aihara
4/2000–9/2002	Research Fellow (DC1) Japan Society for the Promotion of Science, at University of Tokyo, Japan Host researcher: Professor Kazuyuki Aihara

Visiting and Term Appointments

8/2022–	Research Fellow (non-tenured) Center for Computational Social Science, Kobe University, Japan
9/2021–3/2023	Professor (non-tenured) Faculty of Science and Engineering, Waseda University, Japan
4/2020–8/2021	Associate Professor (non-tenured) Faculty of Science and Engineering, Waseda University, Japan
11/2018–8/2019	Turing Fellow Alan Turing Institute, UK (terminated because of NM's move to the USA)
4/2018–3/2021	Visiting Professor Dalian University of Technology, China
10/2008–3/2012	Researcher (part-time appointment) Japan Science and Technology Agency, Basic Research Programs PRESTO, Japan
7/2004–3/2006	Group Leader (part-time appointment) Japan Science and Technology Agency, ERATO Aihara Complexity Modelling Project, Japan

Academic Qualifications

4/2000–9/2002	Ph.D. in Engineering, University of Tokyo, Japan (Department of Mathematical Engineering and Information Physics) Received 9/30/2002 Thesis title: "Duality of Information Coding in Pulse-coupled Neural Networks" Thesis advisor: Professor Kazuyuki Aihara
4/1998–3/2000	M.Sc. in Engineering, University of Tokyo, Japan (Department of Mathematical Engineering and Information Physics) Received 3/29/2000 Thesis title: "Cryptosystems with Discretized Chaotic Maps" Thesis advisor: Professor Kazuyuki Aihara
4/1994–3/1998	B.Sc. in Engineering, University of Tokyo, Japan (Department of Mathematical Engineering and Information Physics) Received 3/27/1998 Thesis title: "Analysis of Time Series with Wavelets" (The thesis was written in Japanese. Translation by NM.) Thesis advisor: Professor Yasunori Okabe
9/2000–6/2001	Academic Year Abroad Department of Physics, University of California, San Diego, USA

Citation Statistics

Google Scholar Citations profile

Grants

University of Michigan, USA

4/2026–3/2031
Japan Science and Technology Agency, Moonshot Research & Development Program, Japan
Award number: JPMJMS2021
Project title: Mathematical modeling and data analysis for understanding and intervening into resilience, robustness, and cascading failures in inter-organ networks
Amount: $1,090,322 (≈ JPY 169,000,000) (to my group)
Role: Co-PI (PI: Kazuyuki Aihara, University of Tokyo)

4/2026–4/2027
Collaborative Minds Mini-Grant Award, Eisenberg Family Depression Center, University of Michigan
[No official subject of the project]
Amount: $10,000
Role: MPI (other MPI: Youngheun Jo; co-Is: Jillian Hardee, Alexander Weigrad, all from University of Michigan)

State University of New York at Buffalo, USA

[Selected for a potential award, but cancelled because transferring the award to a different institution (University of Michigan, to which NM moved on 7/1/2025) would violate DARPA's procurement integrity and solicitation process]
Twelve months from around 5/2025
Defense Advanced Research Projects Agency (DARPA), Advanced Research Concepts (ARC)
Opportunity titled Critical Orientation of Mathematics to Produce Advancements in Science and Security (COMPASS)
Subject: Early warning signals for complex network dynamics
Amount: $297,844
Role: PI (sole PI)

10/2024–9/2025
Research and Economic Development Grant Resubmission Program, State University of New York at Buffalo
Subject: Correlation network analysis: algorithms and application to gene co-expression data
Amount: $15,000
Role: PI (co-I: Omer Gokcumen, State University of New York at Buffalo)

3/2024–12/2024
College of Arts and Sciences proposal incentive program 2023–24 under Category 2, State University of New York at Buffalo
Subject: None
Amount: $5,000
Role: PI (sole PI)

1/2024–12/2024
Institute for Artificial Intelligence and Data Science, State University of New York at Buffalo
Subject: Co-designing an artificial intelligence model with communities to increase treatment success for opioid use disorders
Amount: $50,000
Collaborators: Ekaterina Noyes, Brian Clemency, Henry Taylor, Michael Lamonte (all at State University of New York at Buffalo), Reza Yousefi-Nooraie (University of Rochester); I am the advisor of the hired PhD student (effectively co-advised with Ekaterina Noyes).

9/2022–6/2026
National Institute of General Medical Sciences (NIGMS) and National Science Foundation (administered by NIGMS)

Award number: 1R01GM148973-01
Subject: DMS/NIGMS 1: Multilayer network approach to tandem repeat variation in genomes
Amount: $444,229
Role: PI (co-I: Omer Gokcumen, State University of New York at Buffalo)

[I am only named Senior Personnel] 9/2022–2/2026
National Science Foundation (CCF, NSF 21-590 Predictive Intelligence for Pandemic Prevention Phase I: Development Grants)
Award Number: 2200173
Subject: PIPP Phase I: Center for Ecosystems Data Integration and Pandemic Early Warning Systems
Amount: $1,000,000
Role: Named Senior Personnel (PI: Jennifer Surtees; co-PIs: E. Bruce Pitman, Wen Dong, Laurene Tumiel-Berhalter, Yinyin Ye; all at State University of New York at Buffalo)

9/2022–8/2025
National Science Foundation (Applied Mathematics Program)
Award number: 2204936
Subject: Theory and application of temporal network embedding
Amount: $300,000
Role: PI (sole PI)
Part of this project was moved to University of Michigan alongside my move to Michigan as:
Project period: 1/2026–8/2026
Award number: 2611677
Amount: $85,411

4/2021–3/2024
National Science Foundation (Mathematical Biology Program)
Award number: 2052720
Subject: Floquet theory for stochastic temporal networks and optimization theory for the design of schedules for COVID-19
Amount: $239,983
Role: PI (Co-PI: Dane Taylor, State University of New York at Buffalo)

4/2021–3/2026
Japan Science and Technology Agency, Moonshot Research & Development Program, Japan
Award number: JPMJMS2021
Project title: Mathematical modeling and data analysis for understanding and intervening into resilience, robustness, and cascading failures in inter-organ networks
Amount: $1,618,494 (= JPY 169,000,000) (to my group)
Role: Co-PI (PI: Kazuyuki Aihara, University of Tokyo, Co-PI: Jun-ichi Imura, Tokyo Institute of Technology; Shingo Iwami, Kyushu University; Yukinori Okada, Osaka University; Shigeru Saito, University of Toyama; Kiyohito Kasai, University of Tokyo)
An additional supplement travel grant of JPY 1,758,000 (≈ $ 13,488) was given, 1/2023.
An additional supplement travel grant of JPY 2,730,000 (≈ $ 17,613) was given, 11/2024.

11/2020–11/2021
Sumitomo Foundation
Project title: Analysis of epidemic process models using higher-order random walks on networks
Amount: $13,302.83 (= JPY 1,400,000) (to my group)
Role: PI (Co-I: Gouhei Tanaka, University of Tokyo)

7/2020–6/2021
Nakatani Foundation

Project title: Engineering curfew strategies based on contact network analysis
Amount: $18,497.97 (= JPY 2,000,000) (to my group)
Role: PI (Co-I: Dane Taylor, State University of New York at Buffalo; Naoya Fujiwara, Tohoku University)

5/2020–8/2021
JP Morgan Chase Faculty Research Awards
Project title: Detecting fraudulent transactions in online marketplaces using temporal network motifs
Amount: $150,000
Role: PI (partner PI: A. Erdem Sarıyüce, State University of New York at Buffalo)

4/2020–7/2020
SUNY Research Seed Grant Program
Project title: Optimizing curfew schedules to suppress COVID-19 spreading
Amount: $5,500 ($2,750 to my group)
Role: PI (Other member of the Team: Dane Taylor, State University of New York at Buffalo)

2/2020–8/2023
Air Force Office of Scientific Research (AFOSR) European Office
Award number: FA9550-19-1-7024
Project title: Inferring dynamics of discrete states in time-varying networks
Amount: $144,402 (Subcontracted from University of Bristol, which is NM's previous institution. NM secured $160,709 while he was in Bristol, which has been split between Buffalo and Bristol.)
Role: PI (no Co-PI/Co-I)

Kobe University, Japan (as non-tenured Research Fellow)

4/2024–3/2027
Japan Society for Promotion of Science (Grant-in-Aid for Scientific Research (C))
Award number: 24K14840
Subject: Developing mathematical methods for analyzing correlation networks (translation by NM)
Amount: JPY 4,550,000
Role: PI (no Co-PI/Co-I)

4/2024–3/2028
Japan Society for Promotion of Science (Grant-in-Aid for Scientific Research (B))
Award number: 24K03013
Subject: Revealing mechanisms behind emergence of collective emotion by field sensing (translation by NM)
Amount: JPY 18,590,000 for the entire project (so far JPY 1,430,000 distributed to my group)
Role: Co-I (PI: Ryota Nomura, Waseda University, Co-I: Yutaka Shimada)

4/2023–3/2027
Japan Society for Promotion of Science (Grant-in-Aid for Scientific Research (B))
Award number: 23K28104
Subject: Deriving dynamical model equations from temporal network data using a graph rewriting framework (translation by NM)
Amount: JPY 9,100,000 for the entire project (so far JPY 1,755,000 distributed to my group)
Role: Co-I (PI: Hiroki Sayama, Waseda University)

Waseda University, Japan (as non-tenured faculty member)

4/2021–3/2025
Japan Society for Promotion of Science (Grant-in-Aid for Scientific Research (A))
Award number: 21H04595 Subject: Co-evolution of epidemics, interventions, and behavior in network

epidemiology
Amount: JPY 2,600,000 (to my group)
Role: Co-I (PI: Petter Holme, Tokyo Institute of Technology, Co-Is: Tsuyoshi Murata, Misako Takayasu, Yoko Ibuka, Yusuke Asai, Catherine Beauchemin) [money moved to Kobe University in 4/2024]

University of Bristol, United Kingdom

2/2019–9/2019
GW4 Accelerator Fund
Project title: Recurrence analysis for the characterisation and classification of epileptic patients
Amount: £34,928 (£32,721 to my group)
Role: PI (Co-I: Lorenzo Livi, University of Exeter; Jiaxiang Zhang, Cardiff University; Tiago de Paula Peixoto, University of Bath)

11/2018–8/2019
Alan Turing Institute
Project title: Revealing citation cartels in network data
Amount: £35,139
Role: PI (no Co-PI/Co-I)

9/2018–2/2019
Mercari Inc (a company in Japan)
Project title: Detection of problematic transactions in online marketplaces of Mercari
Amount: £20,371 (company's contribution)
Role: PI (no Co-PI/Co-I)

5/2018–11/2018
EPSRC, BRIM (Building Resilience into Risk Management) Feasibility Fund
Project title: Network resilience theory for water distribution systems
Amount: £10,200
Role: PI (Co-I: Fanlin Meng, University of Exeter)

3/2018–2/2020
Cookpad Limited (a company registered in England and Wales)
Project title: Cookpad networks
Amount: £182,278 (company's contribution)
Role: PI (no Co-PI/Co-I)

3/2018–6/2018
Jean Golding Institute: Seed Corn Funding, University of Bristol
Project title: Immune state networks in wild mice
Amount: £2,600
Role: PI (Co-I: Mark Viney, University of Bristol)

2/2018–6/2018
GW4 Data Science Seed Corn Funding
Project title: Recurrence analysis for time-varying networks and its application to brain dynamics
Amount: £5,000
Role: PI (Co-I: Lorenzo Livi, University of Exeter; Jiaxiang Zhang, Cardiff University)

12/2017–6/2018
Faculty of Engineering: Research Pump Priming, University of Bristol
Project title: Immune response networks in wild mice
Amount: £5,000

Role: PI (Co-I: Mark Viney, University of Bristol)

9/2016–3/2017
EPSRC Institutional Sponsorship
Project title: Phase transitions in neuroimaging data
Amount: £7,234
Role: PI (Co-I: Elohim Fonseca dos Reis, University of Campinas, Brazil)

10/2013–3/2019
CREST, Japan Science and Technology Agency
Project title: Theory for analyzing temporal network data as deep knowledge (originally in Japanese. Translation by NM)
Amount: JPY 62,970,700 (to my group), of which £388,411.21 is to the University of Bristol; the rest is to the University of Tokyo
Role: Co-I (PI: Kenji Yamanishi, University of Tokyo, Japan)

University of Tokyo, Japan

4/2013–2/2014
Bilateral Joint Research Projects, Japan Society for the Promotion of Science
Project title: TempoNet: theoretical foundations of temporal networks
Amount: JPY 2,500,000 (to my group)
Role: PI (partner PI: Renaud Lambiotte, University of Namur, Belgium)

4/2013–3/2014
Research Grants for Japanese Young Scientists (originally in Japanese. Translation by NM), The Nakajima Foundation
Project title: Quantify reputations of agents in networks (originally in Japanese. Translation by NM)
Amount: JPY 1,000,000
Role: PI (no Co-PI/Co-I)

6/2013–8/2013
Japan Society for the Promotion of Science
Grant given to the host of a visiting fellow on JSPS Summer Program
Amount: JPY 100,000
Role: Host of Dr. Friederike Greb

4/2011–2/2014
Japan Society for the Promotion of Science (Grant-in-Aid for Young Scientists (A))
Project title: Computational modeling of games on social networks and analysis of cooperative behavior
Amount: JPY 10,100,000
Role: PI (no Co-PI/Co-I)

2/2012–3/2012
Japan Society for the Promotion of Science
Grant given to the host of a visiting fellow on Postdoctoral Fellowships for Foreign Researchers (short-term) program
Amount: JPY 81,000
Role: Host of Dr. Paul Expert

5/1/2009–2/28/2010
Grant given to the host researcher from the sender institution
Amount: JPY 150,000
Role: Host of Prof. Toshihiro Tanizawa, Kochi National College of Technology, Japan

11/2008–3/2013
Japan Society for the Promotion of Science (Grant-in-Aid for Scientific Research on Innovative Areas)
Project title: Computational study of locomotion, learning, and memory using stochastic analysis methods
Amount: JPY 29,845,000 (to my group)
Role: PI (Co-I: Jun Ohkubo)

10/2008–3/2012
Japan Science and Technology Agency, Basic Research Programs PRESTO, Japan
Project title: Modeling of epidemic dynamics on networks with group structure
Amount: JPY 49,332,600
Role: PI (no Co-PI/Co-I)

4/2008–3/2011
Japan Society for the Promotion of Science (Grant-in-Aid for Young Scientists (B))
Project title: Computational modeling of games on social networks and analysis of cooperative behavior
Amount: JPY 4,030,000
Role: PI (no Co-PI/Co-I)

4/2008–3/2011
Japan Society for Promotion of Science (Grant-in-Aid for Scientific Research (C))
Project title: Renormalization group approach to the analysis of coexistence dynamics on complex networks with randomness (originally in Japanese. Translation by NM)
Amount: JPY 221,000
Role: Co-I (PI: Toshihiro Tanizawa, Kochi National College of Technology)

4/2008–11/2009
Ministry of Education, Culture, Sports, Science and Technology of Japan (Grant-in-Aid for Scientific Research on Priority Areas "Integrative Brain Research")
Project title: Neuro-computational modeling of cooperative behavior (originally in Japanese. Translation by NM)
Amount: JPY 1,787,269
Role: PI (no Co-PI/Co-I)

6/2007–3/2009
Trend Micro Incorporated (a company in Japan)
Project title: Exploration of content security on the WWW using network analysis (originally in Japanese. Translation by NM)
Amount: JPY 5,250,000
Role: PI

4/26/2007–5/26/2007
National Institute for Physiological Sciences, Japan, Japan–U.S. Brain Research Cooperative Program (travel grant)
Project title: A computational study of improvement in behavioral performance by attention
Amount: JPY 570,000
Role: Visiting investigator (Host: Dr. Brent Doiron, Center for Neural Science, New York University)

As postdoc and PhD student in Japan

4/2004–9/2006
Special Postdoctoral Researcher, RIKEN Brain Science Institute, Japan
Project title: Spatiotemporal information processing by recurrent neural networks with synaptic learning and integrate-and-fire dynamics (originally in Japanese. Translation by NM)

Amount: JPY 3,900,000 (research grant portion)
Role: Sole recipient

4/2003–3/2004
Research Fellowships for Young Scientists (PD), Japan Society for the Promotion of Science
Project title: Multiplexity of information representation and contributions of complex spiking patterns and synaptic learning to brain functioning (originally in Japanese. Translation by NM)
Host researcher: Prof. Norio Konno (Yokohama National University, Japan)
Amount: JPY 1,200,000 (research grant portion)
Role: Sole recipient

4/2000–3/2003
Research Fellowships for Young Scientists (DC1), Japan Society for the Promotion of Science
Project title: Analysis of finite-state transformations by chaotic mapping and its application to cryptosystems and neural dynamics (originally in Japanese. Translation by NM)
Host researcher: Prof. Kazuyuki Aihara (University of Tokyo, Japan)
Amount: JPY 3,000,000 (research grant portion)
Role: Sole recipient

Honors and Awards

PLoS ONE, Editorial Board, Long Service Award, 2023

IOP (Institute of Physics) Trusted Reviewer status 2020
(awarded to approximately the top 12% of reviewers since 2018)

JSPS (Japan Society for Promotion of Science) Prize 2020
(awarded to 24 researchers under the age of 45 across all fields of the humanities, social sciences, and natural sciences)

Junior Akira Okubo Prize 2019
jointly administered by the Society for Mathematical Biology and the Japanese Society for Mathematical Biology

Outstanding Reviewer Awards 2017
New Journal of Physics, IOP Publishing (awarded to 82 out of 1,507 reviewers)

3/2016 First Prize for Creating Research (originally in Japanese. Translation by NM)
Research Grants on Brain and Creativity
administered by Neurocreative Laboratory (an NPO in Japan)
Project title: Quantifying drastic changes in brain dynamics (originally in Japanese. Translation by NM) Award prize amount: $10,000

The Young Scientists' Prize
The Commendation for Science and Technology by the Minister of Education, Culture, Sports, Science and Technology, Japan, April 2013.

Japanese Society for Mathematical Biology Early Career Award, 2012.

International Neural Network Society Young Investigator Award
The International Neural Network Society, August 15, 2007.

Refereed Journal Papers

228. Claudia Neuendorf, Naoki Masuda, Kou Murayama.
The global structure of classroom social networks predicts academic and psycho-social adjustment.

PNAS Nexus, in press (2026).

227. Aming Li, Yao Meng, Lei Zhou, <u>Naoki Masuda</u>, Long Wang.
Temporality modulates the effect of network heterogeneity on cooperation fixation.
Nature Communications, 17, 6238 (2026).

226. <u>Naoki Masuda</u>.
Detecting and forecasting tipping points from sample variance alone.
PNAS Nexus, 5, pgag126 (2026).

225. Alber Aqil, Saiful Islam, Faraz Hach, Ibrahim Numanagić, <u>Naoki Masuda</u>, Omer Gokcumen.
Genomes from 117 vertebrate species reveal rapidly evolving segmental duplication landscapes.
Genome Biology and Evolution, 18, evag043 (2026).

224. Shilong Yu, Neil G. MacLaren, <u>Naoki Masuda</u>.
Using covariance of node states to design early warning signals for network dynamics.
Philosophical Transctions of the Royal Society A, 384, 20240486 (2026).

223. Daiki Tatematsu, Naotoshi Nakamura, Masato S. Abe, Tetsuo Ishikawa, Takahiro Ezaki, Lin Cai, Eiryo Kawakami, Kazuyuki Aihara, Atsushi Nishida, Naohiro Okada, <u>Naoki Masuda</u>, Kiyoto Kasai, Shinsuke Koike, Shingo Iwami.
Psychological distress among Japanese high school students during the COVID-19 pandemic: An energy landscape analysis.
PLoS Medicine, 23, e1004884 (2026).

222. Qiao Ke, <u>Naoki Masuda</u>, Zhen Jin, Chuang Liu, Xiu-Xiu Zhan.
Source detection in epidemic dynamics on hypergraphs using a dynamic message passing algorithm.
Chaos, 36, 013121 (2026).

221. Neil G. MacLaren, Baruch Barzel, <u>Naoki Masuda</u>.
Observing network dynamics through sentinel nodes.
Nature Communications, 16, 10211 (2025).

220. <u>Naoki Masuda</u>, Zachary M. Boyd, Diego Garlaschelli, Peter J. Mucha.
Introduction to correlation networks: Interdisciplinary approaches beyond thresholding.
Physics Reports, 1136, 1–39 (2025).

219. Alber Aqil, Yanyan Li, Zhiliang Wang, Saiful Islam, Madison Russel, Theodora Kunovac Kallak, Marie Saitou, Omer Gokcumen, <u>Naoki Masuda</u>.
Switch-like gene expression modulates disease risk.
Nature Communications, 16, 5323 (2025).

218. Lucas Lacasa, F. Javier Marín-Rodríguez, <u>Naoki Masuda</u>, Lluís Arola-Fernández.
Scalar embedding of temporal network trajectories.
Chaos, Solitons and Fractals, 199, 116599 (2025).

217. Chanon Thongprayoon, <u>Naoki Masuda</u>.
Spline tie-decay temporal networks.
Journal of Computational Science, 88, 102591 (2025).

216. <u>Naoki Masuda</u>, Saiful Islam, Si Thu Aung, Takamitsu Watanabe.
Energy landscape analysis based on the Ising model: Tutorial review.
PLoS Complex Systems, 2, e0000039 (2025).

215. Neil G. MacLaren, Kazuyuki Aihara, Naoki Masuda.
Applicability of spatial early warning signals to complex network dynamics.
Journal of the Royal Society Interface, 22, 20240696 (2025).

214. Harrison Hartle, Naoki Masuda.
Autocorrelation properties of temporal networks governed by dynamic node variables.
Physical Review Research, 7, 013083 (2025).

213. Christelle Langley, Giovanni De Marco, Souhir Daly, Naoki Masuda, Angela Davies Smith, Rosemary Jones, Jared Bruce, Ngoc Jade Thai.
Neural substrates of alerting dysfunction in females with multiple sclerosis.
Multiple Sclerosis and Related Disorders, 93, 106208 (2025).

212. Maisha Islam Sejunti, Dane Taylor, Naoki Masuda.
A Parrondo paradox in susceptible-infectious-susceptible dynamics over periodic temporal networks.
Mathematical Biosciences, 378, 109336 (2024).

211. Lucas H. McCabe, Naoki Masuda, Shannon Casillas, Nathan Danneman, Alen Alic, Royal Law.
Network analysis of U.S. non-fatal opioid-involved overdose journeys, 2018–2023.
Applied Network Science, 9, 68 (2024).

210. Hang-Hyun Jo, Tibebe Birhanu, Naoki Masuda.
Temporal scaling theory for bursty time series with clusters of arbitrarily many events.
Chaos, 34, 083110 (2024).

209. Chanon Thongprayoon, Naoki Masuda.
Online landmark replacement for out-of-sample dimensionality reduction methods.
Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 480, 20230966 (2024).

208. Jnanajyoti Bhaumik, Naoki Masuda.
Constant-selection evolutionary dynamics on weighted networks.
Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 480, 20240223 (2024).

207. Katherine Betz, Feng Fu, Naoki Masuda.
Evolutionary game dynamics with environmental feedback in a network with two communities.
Bulletin of Mathematical Biology, 86, 84 (2024).

206. Pitambar Khanra, Johan Nakuci, Sarah Muldoon, Takamitsu Watanabe, Naoki Masuda.
Reliability of energy landscape analysis of resting-state functional MRI data.
European Journal of Neuroscience, 60, 4265–4290 (2024).

205. Ruodan Liu, Naoki Masuda.
Fixation dynamics on multilayer networks.
SIAM Journal on Applied Mathematics, 84, 2028–2050 (2024).

204. Saiful Islam, Pitambar Khanra, Johan Nakuci, Sarah Muldoon, Takamitsu Watanabe, Naoki Masuda.
State-transition dynamics of resting-state functional magnetic resonance imaging data: Model comparison and test-to-retest analysis.
BMC Neuroscience, 25, 14 (2024).

203. Naoki Masuda, Kazuyuki Aihara, Neil G. MacLaren.
Anticipating regime shifts by mixing early warning signals from different nodes.
Nature Communications, 15, 1086 (2024).

202. Jun Miyata, Akihiko Sasamoto, Takahiro Ezaki, Masanori Isobe, Takanori Kochiyama, Naoki Masuda, Yasuo Mori, Yuki Sakai, Nobukatsu Sawamoto, Shisei Tei, Shiho Ubukata, Toshihiko Aso, Toshiya Murai, Hidehiko Takahashi.
Associations of conservatism and jumping to conclusions biases with aberrant salience and default mode network.
Psychiatry and Clinical Neurosciences, 78, 322–331 (2024).

201. Shota Yonezawa, Takayuki Haruki, Keiichi Koizumi, Akinori Taketani, Yusuke Oshima, Makito Oku, Akinori Wada, Tsutomu Sato, Naoki Masuda, Jun Tahara, Noritaka Fujisawa, Shota Koshiyama, Makoto Kadowaki, Isao Kitajima, Shigeru Saito.
Establishing monoclonal gammopathy of undetermined significance as an independent pre-disease state of multiple myeloma using Raman spectroscopy, dynamical network biomarker theory, and energy landscape analysis.
International Journal of Molecular Sciences, 25, 1570 (2024).

200. Neil G. MacLaren, Lingqi Meng, Melissa Collier, Naoki Masuda.
Cooperation and the social brain hypothesis in primate social networks.
Frontiers in Complex Systems, 1, 1344094 (2024).

199. Ruodan Liu, Masaki Ogura, Elohim Fonseca dos Reis, Naoki Masuda.
Effects of concurrency on epidemic spreading in Markovian temporal networks.
European Journal of Applied Mathematics, 35, 430–461 (2024).

198. Ik Soo Lim, Naoki Masuda.
To trust or not to trust: Evolutionary dynamics of an asymmetric N-player trust game.
IEEE Transactions on Evolutionary Computation, 28, 117–131 (2024).

197. Madison Russell, Alber Aqil, Marie Saitou, Omer Gokcumen, Naoki Masuda.
Gene communities in co-expression networks across different tissues.
PLoS Computational Biology, 19, e1011616 (2023).

196. Kashin Sugishita, Naoki Masuda.
Social network analysis of manga: Similarities to real-world social networks and trends over decades.
Applied Network Science, 8, 79 (2023).

195. Kazuki Nakajima, Ruodan Liu, Kazuyuki Shudo, Naoki Masuda.
Quantifying gender imbalance in East Asian academia: Research career and citation practice.
Journal of Informetrics, 17, 101460 (2023).

194. Ruodan Liu, Naoki Masuda.
Fixation dynamics on hypergraphs.
PLoS Computational Biology, 19, e1011494 (2023).

193. Jnanajyoti Bhaumik, Naoki Masuda.
Fixation probability in evolutionary dynamics on switching temporal networks.
Journal of Mathematical Biology, 87, 64 (2023).

192. Maximilian B. Kloucek, Thomas Machon, Shogo Kajimura, C. Patrick Royall, Naoki Masuda, Francesco Turci.
Biases in inverse Ising estimates of near-critical behavior.
Physical Review E, 108, 014109 (2023).

191. Lingqi Meng, Naoki Masuda.
Perturbation theory for evolution of cooperation on networks.
Journal of Mathematical Biology, 87, 12 (2023).

190. Christelle Langley, Naoki Masuda, Simon Godwin, Giovanni De Marco, Angela Davies Smith, Rosemary Jones, Jared Bruce, Ngoc Jade Thai.
Dysfunction of basal ganglia functional connectivity associated with subjective and cognitive fatigue in multiple sclerosis.
Frontiers in Neuroscience, 17, 1194859 (2023).

189. Yiding Cao, Suraj Rajendran, Prathic Sundararajan, Royal K. Law, Sarah Bacon, Steven A. Sumner, Naoki Masuda.
Web-based social networks of individuals with adverse childhood experiences: Quantitative study.
Journal of Medical Internet Research, 25, e45171 (2023).

188. Chanon Thongprayoon, Lorenzo Livi, Naoki Masuda.
Embedding and trajectories of temporal networks.
IEEE Access, 11, 41426–41443 (2023).

187. Neil G. MacLaren, Prosenjit Kundu, Naoki Masuda.
Early warnings for multistage transitions in dynamics on networks.
Journal of the Royal Society Interface, 20, 20220743 (2023).

186. Kazuki Nakajima, Kazuyuki Shudo, Naoki Masuda.
Higher-order rich-club phenomenon in collaborative research grants.
Scientometrics, 128, 2429–2446 (2023).

185. Yukiko Matsumoto, Satoshi Nishida, Ryusuke Hayashi, Shuraku Son, Akio Murakami, Naganobu Yoshikawa, Hiroyoshi Ito, Naoya Oishi, Naoki Masuda, Toshiya Murai, Karl Friston, Shinji Nishimoto, Hidehiko Takahashi.
Disorganization of semantic brain networks in schizophrenia revealed by fMRI.
Schizophrenia Bulletin, 49, 498–506 (2023).

184. Prosenjit Kundu, Neil G. MacLaren, Hiroshi Kori, Naoki Masuda.
Mean-field theory for double-well systems on degree-heterogeneous networks.
Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 478, 20220350 (2022).

    Correction to: 'Mean-field theory for double-well systems on degree-heterogeneous networks' (2023) by Kundu *et al.*.
    Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 479, 20230171 (2023).

183. Takao Sasaki, Naoki Masuda, Richard P. Mann, Dora Biro.
Empirical test of the many-wrongs hypothesis reveals weighted averaging of individual routes in pigeon flocks.
iScience, 25, 105076 (2022).

182. Naoki Masuda, Prosenjit Kundu.
Dimension reduction of dynamical systems on networks with leading and non-leading eigenvectors of adjacency matrices.
Physical Review Research, 4, 023257 (2022).

181. Prosenjit Kundu, Hiroshi Kori, Naoki Masuda.
Accuracy of a one-dimensional reduction of dynamical systems on networks.
Physical Review E, 105, 024305 (2022).

180. Elohim Fonseca dos Reis, Naoki Masuda.
Metapopulation models imply non-Poissonian statistics of interevent times.
Physical Review Research, 4, 013050 (2022).

179. Penghang Liu, Naoki Masuda, Tomomi Kito, Ahmet Erdem Sarıyüce.
Temporal motifs in patent opposition and collaboration networks.
Scientific Reports, 12, 1917 (2022).

178. Kazuki Nakajima, Kazuyuki Shudo, Naoki Masuda.
Randomizing hypergraphs preserving degree correlation and local clustering.
IEEE Transactions on Network Science and Engineering, 9, 1139–1153 (2022).

177. Lingqi Meng, Naoki Masuda.
Epidemic dynamics on metapopulation networks with node2vec mobility.
Journal of Theoretical Biology, 534, 110960 (2022).

176. Marie Saitou, Naoki Masuda, Omer Gokcumen.
Similarity-based analysis of allele frequency distribution among multiple populations identifies adaptive genomic structural variants.
Molecular Biology and Evolution, 39, msab313 (2022).

175. Christos Ellinas, Christos Nicolaides, Naoki Masuda.
Mitigation strategies against cascading failures within a project activity network.
Journal of Computational Social Science, 5, 383–400 (2022).

    Correction to: Mitigation strategies against cascading failures within a project activity network.
    Journal of Computational Social Science, 5, 1097–1098 (2022).

174. Arthur P. C. Spencer, Jonathan C. W. Brooks, Naoki Masuda, Hollie Byrne, Richard Lee-Kelland, Sally Jary, Marianne Thoresen, Marc Goodfellow, Frances M. Cowan, Ela Chakkarapani.
Motor function and white matter connectivity in children cooled for neonatal encephalopathy.
NeuroImage: Clinical, 32, 102872 (2021).

173. Kashin Sugishita, Noha Abdel-Mottaleb, Qiong Zhang, Naoki Masuda.
A growth model for water distribution networks with loops.
Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 477, 20210528 (2021).

172. Takahiro Ezaki, Yu Himeno, Takamitsu Watanabe, Naoki Masuda.
Modeling state-transition dynamics in resting-state brain signals by the hidden Markov and Gaussian mixture models.
European Journal of Neuroscience, 54, 5404–5416 (2021).

171. Hang-Hyun Jo, Naoki Masuda.
Finite-size effects on the convergence time in continuous-opinion dynamics.
Physical Review E, 104, 014309 (2021).

170. Kashin Sugishita, Mason A. Porter, Mariano Beguerisse-Díaz, Naoki Masuda.
Opinion dynamics on tie-decay networks.
Physical Review Research, 3, 023249 (2021).

169. Naoki Masuda, Joel C. Miller, Petter Holme.
Concurrency measures in the era of temporal network epidemiology: A review.
Journal of the Royal Society Interface, 18, 20210019 (2021).

168. Sadamori Kojaku, Giacomo Livan, Naoki Masuda.
Detecting anomalous citation groups in journal networks.
Scientific Reports, 11, 14524 (2021).

167. Kashin Sugishita, Naoki Masuda.
Recurrence in the evolution of air transport networks.
Scientific Reports, 11, 5514 (2021).

166. Arthur P. C. Spencer, Jonathan C. W. Brooks, Naoki Masuda, Hollie Byrne, Richard Lee-Kelland, Sally Jary, Marianne Thoresen, James Tonks, Marc Goodfellow, Frances M. Cowan, Ela Chakkarapani.
Disrupted brain connectivity in children treated with therapeutic hypothermia for neonatal encephalopathy.
NeuroImage: Clinical, 30, 102582 (2021).

165. Elohim Fonseca dos Reis, Mark Viney, Naoki Masuda.
Network analysis of the immune state of mice.
Scientific Reports, 11, 4306 (2021).

164. Marinho A. Lopes, Dominik Krzemiński, Khalid Hamandi, Krish D. Singh, Naoki Masuda, John R. Terry, Jiaxiang Zhang.
A computational biomarker of juvenile myoclonic epilepsy from resting-state MEG.
Clinical Neurophysiology, 132, 922–927 (2021).

163. Marinho A. Lopes, Jiaxiang Zhang, Dominik Krzemiński, Khalid Hamandi, Qi Chen, Lorenzo Livi, Naoki Masuda.
Recurrence quantification analysis of dynamic brain networks.
European Journal of Neuroscience, 53, 1040–1059 (2021).

162. Lingqi Meng, Naoki Masuda.
Analysis of node2vec random walks on networks.
Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 476, 20200447 (2020).

161. Shun Kodate, Ryusuke Chiba, Shunya Kimura, Naoki Masuda.
Detecting problematic transactions in a consumer-to-consumer e-commerce network.
Applied Network Science, 5, 90 (2020).

160. Elohim Fonseca dos Reis, Aming Li, Naoki Masuda.
Generative models of simultaneously heavy-tailed distributions of inter-event times on nodes and edges.
Physical Review E, 102, 052303 (2020).

159. Shogo Kajimura, Naoki Masuda, Johnny King L. Lau, Kou Murayama.
Focused attention meditation changes the boundary and configuration of functional networks in the brain.
Scientific Reports, 10, 18426 (2020).

158. Xiu-Xiu Zhan, Ziyu Li, Naoki Masuda, Petter Holme, Huijuan Wang.
Susceptible-infected-spreading-based network embedding in static and temporal networks.
EPJ Data Science, 9, 30 (2020).

157. Mengqiao Xu, Qian Pan, Haoxiang Xia, Naoki Masuda.
Estimating international trade status of countries from global liner shipping networks.
Royal Society Open Science, 7, 200386 (2020).

156. Irene Malvestio, Alessio Cardillo, Naoki Masuda.
Interplay between k-core and community structure in complex networks.
Scientific Reports, 10, 14702 (2020).

155. Genki Ichinose, Daiki Miyagawa, Junji Ito, Naoki Masuda.
Winning by hiding behind others: An analysis of speed skating data.
PLoS ONE, 15, e0237470 (2020).

154. Naoki Masuda, Takayuki Hiraoka.
Waiting-time paradox in 1922.
Northeast Journal of Complex Systems, 2, 1 (2020).

153. Alessio Cardillo, Naoki Masuda.
Critical mass effect in evolutionary games triggered by zealots.
Physical Review Research, 2, 023305 (2020).

152. Naoki Masuda, Petter Holme.
Small inter-event times govern epidemic spreading on networks.
Physical Review Research, 2, 023163 (2020).

151. Takayuki Hiraoka, Naoki Masuda, Aming Li, Hang-Hyun Jo.
Modeling temporal networks with bursty activity patterns of nodes and links.
Physical Review Research, 2, 023073 (2020).

150. Gabriele Valentini, Naoki Masuda, Zachary Shaffer, Jake R. Hanson, Takao Sasaki, Sara Imari Walker, Theodore P. Pavlic, Stephen C. Pratt.
Division of labor promotes the spread of information in colony emigrations by the ant *Temnothorax rugatulus*.
Proceedings of the Royal Society B: Biological Sciences, 287, 20192950 (2020).

149. Makoto Okada, Kenji Yamanishi, Naoki Masuda.
Long-tailed distributions of inter-event times as mixtures of exponential distributions.
Royal Society Open Science, 7, 191643 (2020).

148. Naoki Masuda, Victor M. Preciado, Masaki Ogura.
Analysis of the susceptible-infected-susceptible epidemic dynamics in networks via the non-backtracking matrix.
IMA Journal of Applied Mathematics, 85, 214–230 (2020).

147. Takahiro Ezaki, Elohim Fonseca dos Reis, Takamitsu Watanabe, Michiko Sakaki, Naoki Masuda.
Closer to critical resting-state neural dynamics in individuals with higher fluid intelligence.
Communications Biology, 3, 52 (2020).

146. Dominik Krzemiński, Naoki Masuda, Khalid Hamandi, Krish D. Singh, Bethany Routley, Jiaxiang Zhang.
Energy landscape of resting magnetoencephalography reveals fronto-parietal network impairments in epilepsy.
Network Neuroscience, 4, 374–396 (2020).

145. Sadamori Kojaku, Naoki Masuda.
Constructing networks by filtering correlation matrices: A null model approach.
Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 475, 20190578 (2019).

144. Naoki Masuda, Fanlin Meng.
Dynamical stability of water distribution networks.
Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 475, 20190291 (2019).

143. Masaki Ogura, Victor M. Preciado, Naoki Masuda.
Optimal containment of epidemics over temporal activity-driven networks.
SIAM Journal on Applied Mathematics, 79, 986–1006 (2019).

142. Takamitsu Watanabe, Geraint Rees, Naoki Masuda.
Atypical intrinsic neural timescale in autism.
eLife, 8, e42256 (2019).

141. Naoki Masuda, Petter Holme.
Detecting sequences of system states in temporal networks.
Scientific Reports, 9, 795 (2019).

140. Sadamori Kojaku, Mengqiao Xu, Haoxiang Xia, Naoki Masuda.
Multiscale core-periphery structure in a global liner shipping network.
Scientific Reports, 9, 404 (2019).

139. Naoki Masuda, Sadamori Kojaku, Yukie Sano.
Configuration model for correlation matrices preserving the node strength.
Physical Review E, 98, 012312 (2018).

138. Sadamori Kojaku, Giulio Cimini, Guido Caldarelli, Naoki Masuda.
Structural changes in the interbank market across the financial crisis from multiple core-periphery analysis.
Journal of Network Theory in Finance, 4, 33–51 (2018).

137. Takuya Sekiguchi, Kohei Tamura, Naoki Masuda.
Population changes in residential clusters in Japan.
PLoS ONE, 13, e0197144 (2018).

136. Sadamori Kojaku, Naoki Masuda.
A generalised significance test for individual communities in networks.
Scientific Reports, 8, 7351 (2018).

135. Sadamori Kojaku, Naoki Masuda.
Core-periphery structure requires something else in the network.
New Journal of Physics, 20, 043012 (2018).

134. Takahiro Ezaki, Michiko Sakaki, Takamitsu Watanabe, Naoki Masuda.
Age-related changes in the ease of dynamical transitions in human brain activity.
Human Brain Mapping, 39, 2673–2688 (2018).

133. Naoki Masuda, Michiko Sakaki, Takahiro Ezaki, Takamitsu Watanabe.
Clustering coefficients for correlation networks.
Frontiers in Neuroinformatics, 12, 7 (2018).

132. Genki Ichinose, Naoki Masuda.
Zero-determinant strategies in finitely repeated games.
Journal of Theoretical Biology, 438, 61–77 (2018).

131. Naoki Masuda, Luis E. C. Rocha.
A Gillespie algorithm for non-Markovian stochastic processes.
SIAM Review, 60, 95–115 (2018).

130. Takahiro Ezaki, Naoki Masuda.
Reinforcement learning account of network reciprocity.
PLoS ONE, 12, e0189220 (2017).

129. Sadamori Kojaku, Naoki Masuda.
Finding multiple core-periphery pairs in networks.
Physical Review E, 96, 052313 (2017).

128. Luis E. C. Rocha, Naoki Masuda, Petter Holme.
Sampling of temporal networks: Methods and biases.
Physical Review E, 96, 052302 (2017).

127. Tomokatsu Onaga, James P. Gleeson, Naoki Masuda.
Concurrency-induced transitions in epidemic dynamics on temporal networks.
Physical Review Letters, 119, 108301 (2017).

126. Naoki Masuda, Mason A. Porter, Renaud Lambiotte.
Random walks and diffusion on networks.
Physics Reports, 716–717, 1–58 (2017).

    Corrigendum to "Random walks and diffusion on networks" [Phys. Rep. 716-717 (2017) 1-58].
    Physics Reports, 745, 96 (2018).

    Second corrigendum to "Random walks and diffusion on networks" [Phys. Rep. 716-717 (2017) 1-58].
    Physics Reports, 851, 37-39 (2020).

125. Kohei Tamura, Naoki Masuda.
Effects of the distant population density on spatial patterns of demographic dynamics.
Royal Society Open Science, 4, 170391 (2017).

124. Takahiro Ezaki, Takamitsu Watanabe, Masayuki Ohzeki, Naoki Masuda.
Energy landscape analysis of neuroimaging data.
Philosophical Transactions of the Royal Society A, 375, 20160287 (2017).

123. Thomas A. O'Shea-Wheller, Naoki Masuda, Ana Sendova-Franks, Nigel R. Franks.
Variability in individual assessment behaviour and its implications for collective decision-making.
Proceedings of the Royal Society B: Biological Sciences, 284, 20162237 (2017).

122. Yutaka Horita, Masanori Takezawa, Keigo Inukai, Toshimasa Kita, Naoki Masuda.
Reinforcement learning accounts for moody conditional cooperation behavior: experimental results.
Scientific Reports, 7, 39275 (2017).

121. Teruyoshi Kobayashi, Naoki Masuda.
Fragmenting networks by targeting collective influencers at a mesoscopic level.
Scientific Reports, 6, 37778 (2016).

120. Luis E. C. Rocha, Naoki Masuda.
Individual-based approach to epidemic processes on arbitrary dynamic contact networks.
Scientific Reports, 6, 31456 (2016).

119. Takahiro Ezaki, Yutaka Horita, Masanori Takezawa, Naoki Masuda.
Reinforcement learning explains conditional cooperation and its moody cousin.
PLoS Computational Biology, 12, e1005034 (2016).

118. Leo Speidel, Konstantin Klemm, Víctor M. Eguíluz, Naoki Masuda.
Temporal interactions facilitate endemicity in the susceptible-infected-susceptible epidemic model.
New Journal of Physics, 18, 073013 (2016).

117. Ryosuke Nishi, Taro Takaguchi, Keigo Oka, Takanori Maehara, Masashi Toyoda, Ken-ichi Kawarabayashi, Naoki Masuda.
Reply trees in Twitter: data analysis and branching process models.
Social Network Analysis and Mining, 6, 26 (2016).

116. Naoki Masuda.
Accelerating coordination in temporal networks by engineering the link order.
Scientific Reports, 6, 22105 (2016).

115. Yutaka Horita, Masanori Takezawa, Takuji Kinjo, Yo Nakawake, Naoki Masuda.
Transient nature of cooperation by pay-it-forward reciprocity.
Scientific Reports, 6, 19471 (2016).

114. Jens Malmros, Naoki Masuda, Tom Britton.
Random walks on directed networks: Inference and respondent-driven sampling.
Journal of Official Statistics, 32, 433–459 (2016).

113. Leo Speidel, Taro Takaguchi, Naoki Masuda.
Community detection in directed acyclic graphs.
European Physical Journal B, 88, 203 (2015).

112. Kohei Tamura, Naoki Masuda.
Win-stay lose-shift strategy in formation changes in football.
EPJ Data Science, 4, 9 (2015).

111. Nigel Franks, Jonathan Stuttard, Carolina Doran, Julian Esposito, Maximillian Master, Ana Sendova-Franks, Naoki Masuda, Nicholas Britton.
How ants use quorum sensing to estimate the average quality of a fluctuating resource.
Scientific Reports, 5, 11890 (2015).

110. Naoki Masuda, Thomas A. O'Shea-Wheller, Carolina Doran, Nigel R. Franks.
Computational model of collective nest selection by ants with heterogeneous acceptance thresholds.
Royal Society Open Science, 2, 140533 (2015).

109. Naoki Masuda, Feng Fu.
Evolutionary models of in-group favoritism.
F1000Prime Reports, 7, 27 (2015).

108. Naoki Masuda.
Opinion control in complex networks.
New Journal of Physics, 17, 033031 (2015).

107. Petter Holme, Naoki Masuda.
The basic reproduction number as a predictor for epidemic outbreaks in temporal networks.
PLoS ONE, 10, e0120567 (2015).

106. Víctor M. Eguíluz, Naoki Masuda, Juan Fernández-Gracia.
Bayesian decision making in human collectives with binary choices.
PLoS ONE, 10, e0121332 (2015).

105. Leo Speidel, Renaud Lambiotte, Kazuyuki Aihara, Naoki Masuda.
Steady state and mean recurrence time for random walks on stochastic temporal networks.
Physical Review E, 91, 012806 (2015).

104. Yohei Nakajima, Naoki Masuda.
Evolutionary dynamics in finite populations with zealots.
Journal of Mathematical Biology, 70, 465–484 (2015).

103. Takamitsu Watanabe, Naoki Masuda, Fukuda Megumi, Ryota Kanai, Geraint Rees.
Energy landscape and dynamics of brain activity during human bistable perception.
Nature Communications, 5, 4765 (2014).

102. Hiroyuki Shimoji, Masato S. Abe, Kazuki Tsuji, Naoki Masuda.
Global network structure of dominance hierarchy of ant workers.
Journal of the Royal Society Interface, 11, 20140599 (2014).

101. Ryosuke Nishi, Naoki Masuda.
Dynamics of social balance under temporal interaction.
Europhysics Letters, 107, 48003 (2014).

100. Yuni Iwamasa, Naoki Masuda.
Networks maximizing the consensus time of voter models.
Physical Review E, 90, 012816 (2014).

99. Naoki Masuda.
Voter model on the two-clique graph.
Physical Review E, 90, 012802 (2014).

98. Takamitsu Watanabe, Shigeyuki Kan, Takahiko Koike, Masaya Misaki, Seiki Konishi, Satoru Miyauchi, Yasushi Miyashita, Naoki Masuda.
Network-dependent modulation of brain activity during sleep.
NeuroImage, 98, 1–10 (2014).

97. Luis E. C. Rocha, Naoki Masuda.
Random walk centrality for temporal networks.
New Journal of Physics, 16, 063023 (2014).

96. Takamitsu Watanabe, Masanori Takezawa, Yo Nakawake, Akira Kunimatsu, Hidenori Yamasue, Mitsuhiro Nakamura, Yasushi Miyashita, Naoki Masuda.
Two distinct neural mechanisms underlying indirect reciprocity.
Proceedings of the National Academy of Sciences of the United States of America, 111, 3990–3995 (2014).

95. Naoki Masuda.
Evolution via imitation among like-minded individuals.
Journal of Theoretical Biology, 349, 100–108 (2014).

94. Koji Oishi, Manuel Cebrian, Andres Abeliuk, Naoki Masuda.
Iterated crowdsourcing dilemma game.
Scientific Reports, 4, 4100 (2014).

93. Takamitsu Watanabe, Satoshi Hirose, Hiroyuki Wada, Yoshio Imai, Toru Machida, Ichiro Shirouzu, Seiki Konishi, Yasushi Miyashita, Naoki Masuda.
Energy landscapes of resting-state brain networks.
Frontiers in Neuroinformatics, 8, 12 (2014).

92. Kodai Saito, Naoki Masuda.
Two types of well followed users in the followership networks of Twitter.
PLoS ONE, 9, e84265 (2014).

91. Shoma Tanabe, Naoki Masuda.
Complex dynamics of a nonlinear voter model with contrarian agents.
Chaos, 23, 043136 (2013).

90. Naoki Masuda, Konstantin Klemm, Víctor M. Eguíluz.
Temporal networks: slowing down diffusion by long lasting interactions.
Physical Review Letters, 111, 188701 (2013).

89. Naoki Masuda.
Voter models with contrarian agents.
Physical Review E, 88, 052803 (2013).

88. Takehisa Hasegawa, Taro Takaguchi, Naoki Masuda.
Observability transitions in correlated networks.
Physical Review E, 88, 042809 (2013).

87. Makoto Hiroi, Masamichi Ohkura, Junichi Nakai, Naoki Masuda, Koichi Hashimoto, Kiichi Inoue, André Fiala, Tetsuya Tabata.
Principal component analysis of odor coding at the level of third order olfactory neurons in *Drosophila*.
Genes to Cells, 18, 1070–1081 (2013).

86. Ryosuke Nishi, Naoki Masuda.
Collective opinion formation model under Bayesian updating and confirmation bias.
Physical Review E, 87, 062123 (2013).

85. Taro Takaguchi, Naoki Masuda, Petter Holme.
Bursty communication patterns facilitate spreading in a threshold-based epidemic dynamics.
PLoS ONE, 8, e68629 (2013).

84. Naoki Masuda, Issei Kurahashi, Hiroko Onari.
Suicide ideation of individuals in online social networks.
PLoS ONE, 8, e62262 (2013).

    Correction: Suicide ideation of individuals in online social networks.
    PLoS ONE, 9, 10.1371/annotation/d589857d-b3c6-4a16-acfe-423f9bf529f1 (2014).

83. Masayoshi Ito, Naoki Masuda, Kazunori Shinomiya, Keita Endo, Kei Ito.
Systematic analysis of neural projections reveals clonal composition of the *Drosophila* brain.
Current Biology, 23, 644–655 (2013).

82. Naoki Masuda, Petter Holme.
Predicting and controlling infectious disease epidemics using temporal networks.
F1000Prime Reports, 5, 6 (2013).

81. Shun-ichi Amari, Hiroyasu Ando, Taro Toyoizumi, Naoki Masuda.
State concentration exponent as a measure of quickness in Kauffman-type networks.
Physical Review E, 87, 022814 (2013).

80. Takamitsu Watanabe, Satoshi Hirose, Hiroyuki Wada, Yoshio Imai, Toru Machida, Ichiro Shirouzu, Seiki Konishi, Yasushi Miyashita, Naoki Masuda.
A pairwise maximum entropy model accurately describes resting-state human brain networks.
Nature Communications, 4, 1370 (2013).

79. Ryo Fujie, Kazuyuki Aihara, Naoki Masuda.
A model of competition among more than two languages.
Journal of Statistical Physics, 151, 289–303 (2013).

78. Shoma Tanabe, Hideyuki Suzuki, Naoki Masuda.
Indirect reciprocity with trinary reputations.
Journal of Theoretical Biology, 317, 338–347 (2013).

    Corrigendum to "Indirect reciprocity with trinary reputations" [J. Theor. Biol. 317 (2013) 338–347].
    Journal of Theoretical Biology, 328, 100–101 (2013).

77. Mitsuhiro Nakamura, Naoki Masuda.
Groupwise information sharing promotes ingroup favoritism in indirect reciprocity.
BMC Evolutionary Biology, 12, 213 (2012).

76. Shun Motegi, Naoki Masuda.
A network-based dynamical ranking system for competitive sports.
Scientific Reports, 2, 904 (2012).

75. Naoki Masuda.
Evolution of cooperation driven by zealots.
Scientific Reports, 2, 646 (2012).

74. Taro Takaguchi, Nobuo Sato, Kazuo Yano, Naoki Masuda.
Importance of individual events in temporal networks.
New Journal of Physics, 14, 093003 (2012).

73. Naoki Masuda, Mitsuhiro Nakamura.
Coevolution of trustful buyers and cooperative sellers in the trust game.
PLoS ONE, 7, e44169 (2012).

72. Naoki Masuda.
Ingroup favoritism and intergroup cooperation under indirect reciprocity based on group reputation.
Journal of Theoretical Biology, 311, 8–18 (2012).

71. Taro Ueno, Naoki Masuda, Shoen Kume, Kazuhiko Kume.
Dopamine modulates the rest period length without perturbation of its power law distribution in *Drosophila melanogaster*.
PLoS ONE, 7, e32007 (2012).

70. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Structure of cell networks critically determines oscillation regularity.
Journal of Theoretical Biology, 297, 61–72 (2012).

69. Shoma Tanabe, Naoki Masuda.
Evolution of cooperation facilitated by reinforcement learning with adaptive aspiration levels.
Journal of Theoretical Biology, 293, 151–160 (2012).

68. Naoki Masuda.
Clustering in large networks does not promote upstream reciprocity.
PLoS ONE, 6, e25190 (2011).

67. Taro Takaguchi, Naoki Masuda.
Voter model with non-Poissonian interevent intervals.
Physical Review E, 84, 036115 (2011).

66. Takehisa Hasegawa, Naoki Masuda.
Robustness of networks against propagating attacks under vaccination strategies.
Journal of Statistical Mechanics, 2011, P09014 (2011).

65. Taro Takaguchi, Mitsuhiro Nakamura, Nobuo Sato, Kazuo Yano, Naoki Masuda.
Predictability of conversation partners.
Physical Review X, 1, 011008 (2011).

64. Mitsuhiro Nakamura, Naoki Masuda.
Indirect reciprocity under incomplete observation.
PLoS Computational Biology, 7, e1002113 (2011).

63. C. -K. Yun, N. Masuda, B. Kahng.
Diversity and critical behavior in prisoner's dilemma game.
Physical Review E, 83, 057102 (2011).

62. Naoki Masuda, Mitsuhiro Nakamura.
Numerical analysis of a reinforcement learning model with the dynamic aspiration level in the iterated Prisoner's Dilemma.
Journal of Theoretical Biology, 278, 55–62 (2011).

61. Takehisa Hasegawa, Norio Konno, Naoki Masuda.
Numerical study of a three-state host-parasite system on the square lattice.
Physical Review E, 83, 046102 (2011).

60. Yuri Ogiso, Kazuhide Tsuneizumi, Naoki Masuda, Makoto Sato, Tetsuya Tabata.
Robustness of the Dpp morphogen activity gradient depends on negative feedback regulation by the inhibitory Smad, Dad.
Development Growth and Differentiation, 53, 668–678 (2011).

59. Naoki Masuda, S. Redner.
Can partisan voting lead to truth?
Journal of Statistical Mechanics, 2011, L02002 (2011).

58. Naoki Masuda, Hiroshi Kori.
Dynamics-based centrality for directed networks.
Physical Review E, 82, 056107 (2010).

57. Takamitsu Watanabe, Naoki Masuda.
Enhancing the spectral gap of networks by node removal.
Physical Review E, 82, 046102 (2010).

56. Jun Ohkubo, Kazushi Yoshida, Yuichi Iino, Naoki Masuda.
Long-tail behavior in locomotion of *Caenorhabditis* elegans.
Journal of Theoretical Biology, 267, 213–222 (2010).

55. Naoki Masuda.
Effects of diffusion rates on epidemic spreads in metapopulation networks.
New Journal of Physics, 12, 093009 (2010).

54. Naoki Masuda, Yoji Kawamura, Hiroshi Kori.
Collective fluctuations in networks of noisy components.
New Journal of Physics, 12, 093007 (2010).

53. Ralf Tönjes, Naoki Masuda, Hiroshi Kori.
Synchronization transition of identical phase oscillators in a directed small-world network.
Chaos, 20, 033108 (2010).

52. Naoki Masuda, N. Gibert, S. Redner.
Heterogeneous voter models.
Physical Review E, 82, 010103(R) (2010).

51. Akio Iwagami, Naoki Masuda.
Upstream reciprocity in heterogeneous networks.
Journal of Theoretical Biology, 265, 297–305 (2010).

50. Yusuke Ide, Norio Konno, Naoki Masuda.
Statistical properties of a generalized threshold network model.
Methodology & Computing in Applied Probability, 12, 361–377 (2010).

49. Naoki Masuda.
Immunization of networks with community structure.
New Journal of Physics, 11, 123018 (2009).

48. Naoki Masuda, Yoji Kawamura, Hiroshi Kori.
Impact of hierarchical modular structure on ranking of individual nodes in directed networks.
New Journal of Physics, 11, 113002 (2009).

47. Naoki Masuda, Yoji Kawamura, Hiroshi Kori.
Analysis of relative influence of nodes in directed networks.
Physical Review E, 80, 046114 (2009).

46. Naoki Masuda, Hisashi Ohtsuki.
A theoretical analysis of temporal difference learning in the iterated Prisoner's Dilemma game.
Bulletin of Mathematical Biology, 71, 1818–1850 (2009).

45. Naoki Masuda.
Selective population rate coding: a possible computational role of gamma oscillations in selective attention.
Neural Computation, 21, 3335–3362 (2009).

44. Yuko K. Takahashi, Hiroshi Kori, Naoki Masuda.
Self-organization of feedforward structure and entrainment in excitatory neural networks with spike-timing-dependent plasticity.
Physical Review E, 79, 051904 (2009).

43. Naoki Masuda, Hisashi Ohtsuki.
Evolutionary dynamics and fixation probabilities in directed networks.
New Journal of Physics, 11, 033012 (2009).

42. Naoki Masuda.
Directionality of contact networks suppresses selection pressure in evolutionary dynamics.
Journal of Theoretical Biology, 258, 323–334 (2009).

41. N. Masuda, J. S. Kim, B. Kahng.
Priority queues with bursty arrivals of incoming tasks.
Physical Review E, 79, 036106 (2009).

40. Taro Ueno, Naoki Masuda.
Controlling nosocomial infection based on structure of hospital social networks.
Journal of Theoretical Biology, 254, 655–666 (2008).

39. Naoki Masuda.
Oscillatory dynamics in evolutionary games are suppressed by heterogeneous adaptation rates of players.
Journal of Theoretical Biology, 251, 181–189 (2008).

38. Naoki Masuda, Shun-ichi Amari.
A computational study of synaptic mechanisms of partial memory transfer in cerebellar vestibulo-ocular-reflex learning.
Journal of Computational Neuroscience, 24, 137–156 (2008).

37. Nobuaki Sugimine, Naoki Masuda, Norio Konno, Kazuyuki Aihara.
On global and local critical points of extended contact process on homogeneous trees.
Mathematical Biosciences, 213, 13–17 (2008).

36. Naoki Masuda, Brent Doiron.
Gamma oscillations of spiking neural populations enhance signal discrimination.
PLoS Computational Biology, 3, e236, 2348–2355 (2007).

35. Naoki Masuda.
Participation costs dismiss the advantage of heterogeneous networks in evolution of cooperation.
Proceedings of the Royal Society B: Biological Sciences, 274, 1815–1821 (2007).

34. Naoki Masuda, Hiroshi Kori.
Formation of feedforward networks and frequency synchrony by spike-timing-dependent plasticity.
Journal of Computational Neuroscience, 22, 327–345 (2007).

33. Naoki Masuda, Masato Okada, Kazuyuki Aihara.
Filtering of spatial bias and noise inputs by spatially structured neural networks.
Neural Computation, 19, 1854–1870 (2007).

32. Naoki Masuda, Kazuyuki Aihara.
Dual coding hypotheses for neural information representation.
Mathematical Biosciences, 207, 312–321 (2007).

31. Naoki Masuda, Hisashi Ohtsuki.
Tag-based indirect reciprocity by incomplete social information.
Proceedings of the Royal Society B: Biological Sciences, 274, 689–695 (2007).

30. Yong-Yeol Ahn, Hawoong Jeong, Naoki Masuda, Jae Dong Noh.
Epidemic dynamics of two species of interacting particles on scale-free networks.
Physical Review E, 74, 066113 (2006).

29. Naoki Masuda, Norio Konno.
Networks with dispersed degrees save stable coexistence of species in cyclic competition.
Physical Review E, 74, 066102 (2006).

28. Yuichi Katori, Naoki Masuda, Kazuyuki Aihara.
Dynamic switching of optimal neural codes in networks with gap junctions.
Neural Networks, 19, 1463–1466 (2006).

27. Naoki Masuda, Norio Konno.
Multi-state epidemic processes on complex networks.
Journal of Theoretical Biology, 243, 64–75 (2006).

26. Kazumichi Ohtsuka, Norio Konno, Naoki Masuda, Kazuyuki Aihara.
Phase diagrams and correlation inequalities of a three-state stochastic epidemic model on the square lattice.
International Journal of Bifurcation and Chaos, 16, 3687–3693 (2006).

25. Naoki Masuda, Norio Konno.
VIP-club phenomenon: emergence of elites and masterminds in social networks.
Social Networks, 28, 297–309 (2006).

24. Naoki Masuda, Goce Jakimoski, Kazuyuki Aihara, Ljupco Kocarev.
Chaotic block ciphers: from theory to practical algorithms.
IEEE Transactions on Circuits and Systems Part I, 53, 1341–1352 (2006).

23. Naoki Masuda.
Simultaneous rate-synchrony codes in populations of spiking neurons.
Neural Computation, 18, 45–59 (2006).

22. N. Masuda, K.-I. Goh, B. Kahng.
Extremal dynamics on complex networks: Analytic solutions.
Physical Review E, 72, 066106 (2005).

21. Norio Konno, Naoki Masuda, Rahul Roy, Anish Sarkar.
Rigorous results on the threshold network model.
Journal of Physics A: Mathematical and General, 38, 6277–6291 (2005).

20. Naoki Masuda, Brent Doiron, André Longtin, Kazuyuki Aihara.
Coding of temporally varying signals in networks of spiking neurons with global delayed feedback.
Neural Computation, 17, 2139–2175 (2005).

19. Naoki Masuda, Hiroyoshi Miwa, Norio Konno.
Geographical threshold graphs with small-world and scale-free properties.
Physical Review E, 71, 036108 (2005).

18. Naoki Masuda, Hiroyoshi Miwa, Norio Konno.
Analysis of scale-free networks based on a threshold graph with intrinsic vertex weights.
Physical Review E, 70, 036124 (2004).

17. Naoki Masuda, Norio Konno.
Subcritical behavior in the alternating supercritical Domany-Kinzel dynamics.
European Physical Journal B, 40, 313–319 (2004).

16. Naoki Masuda, Norio Konno.
Return times of random walk on generalized random graphs.
Physical Review E, 69, 066113 (2004).

15. Naoki Masuda, Kazuyuki Aihara.
Dual coding and effects of global feedback in multilayered neural networks.
Neurocomputing, 58–60, 33–39 (2004).

14. Naoki Masuda, Norio Konno, Kazuyuki Aihara.
Transmission of severe acute respiratory syndrome in dynamical small-world networks.
Physical Review E, 69, 031917 (2004).

13. Naoki Masuda, Kazuyuki Aihara.
Self-organizing dual coding based on spike-time-dependent plasticity.
Neural Computation, 16, 627–663 (2004).

12. Naoki Masuda, Kazuyuki Aihara.
Global and local synchrony of coupled neurons in small-world networks.
Biological Cybernetics, 90, 302–309 (2004).

11. Naoki Masuda, Kazuyuki Aihara.
Spatial prisoner's dilemma optimally played in small-world networks.
Physics Letters A, 313, 55–61 (2003).

10. Naoki Masuda, Kazuyuki Aihara.
Filtered interspike interval encoding by class II neurons.
Physics Letters A, 311, 485–490 (2003).

9. Naoki Masuda, Kazuyuki Aihara.
Ergodicity of spike trains: when does trial averaging make sense?
Neural Computation, 15, 1341–1372 (2003).

8. Naoki Masuda, Kazuyuki Aihara.
Duality of rate coding and temporal spike coding in multilayered feedforward networks.
Neural Computation, 15, 103–125 (2003).

7. Naoki Masuda, Kazuyuki Aihara.
Bridging rate coding and temporal spike coding by effect of noise.
Physical Review Letters, 88, 248101 (2002).

6. Naoki Masuda, Kazuyuki Aihara.
Spatiotemporal spike encoding of a continuous external signal.
Neural Computation, 14, 1599–1628 (2002).

5. Naoki Masuda, Kazuyuki Aihara.
Dynamical characteristics of discretized chaotic permutations.
International Journal of Bifurcation and Chaos, 12, 2087–2103 (2002).

4. Naoki Masuda, Kazuyuki Aihara.
Cryptosystems with discretized chaotic maps.
IEEE Transactions on Circuits and Systems Part I, 49, 28–40 (2002).

3. Naoki Masuda, Kazuyuki Aihara.
Synchronization of pulse-coupled excitable neurons.
Physical Review E, 64, 051906 (2001).

2. Henry D. I. Abarbanel, Naoki Masuda, M. I. Rabinovich, Evren Tumer.
   Distribution of mutual information.
   Physics Letters A, 281, 368–373 (2001).

1. Naoki Masuda, Yasunori Okabe.
   Time series analysis with wavelet coefficients.
   Japan Journal of Industrial and Applied Mathematics, 18, 129–158 (2001).

Editorial

1. Teruyoshi Kobayashi, Naoki Masuda.
   Introduction to the special issue "Economics and Complex Networks".
   Japanese Economic Review, 72, 1–4 (2021).

Commentary

2. Naoki Masuda.
   Predicting tipping points in complex systems.
   (American Physical Society) Viewpoint (Physics), 17, 110 (2024).

   This is a viewpoint article for: Zijia Liu, Xiaozhu Zhang, Xiaolei Ru, Ting-Ting Gao, Jack Murdoch Moore, Gang Yan.
   Early predictor for the onset of critical transitions in networked dynamical systems.
   Physical Review X, 14, 031009 (2024).

1. Naoki Masuda, Francisco C. Santos.
   A mathematical look at empathy.
   eLife, 8, e47036 (2019).

Articles for Young Adults (Refereed)

1. Naoki Masuda, Mason A. Porter.
   The waiting-time paradox.
   Frontiers for Young Minds, 8, 582433 (2021).

Books

4. Naoki Masuda, Christian L. Vestergaard.
   Gillespie Algorithms for Stochastic Multiagent Dynamics in Populations and Networks.
   Cambridge Elements, Cambridge University Press, Cambridge, UK (2022).

3. Naoki Masuda, Renaud Lambiotte.
   A Guide to Temporal Networks (second edition).
   World Scientific, Singapore (2020).
   (The first edition was published in 2016.)

2. Naoki Masuda, Kwang-Il Goh, Tao Jia, Junichi Yamanoi, Hiroki Sayama. (Editors)
   Proceedings of NetSci-X 2020: Sixth International Winter School and Conference on Network Science.
   Springer, Cham, Switzerland (2020).

1. Naoki Masuda, Petter Holme. (Editors)
Temporal Network Epidemiology.
Springer, Singapore (2017).

Book Chapters

4. Tomokatsu Onaga, James P. Gleeson, Naoki Masuda.
The effect of concurrency on epidemic threshold in time-varying networks.
In: Temporal Network Theory, Petter Holme and Jari Saramäki (Eds.), Springer, Cham, Switzerland (2019). pp. 253–267.
The same chapter is in pp. 259–274 of Second Edition (2023).

3. Naoki Masuda, Petter Holme.
Introduction to temporal network epidemiology.
In: Temporal Network Epidemiology, Naoki Masuda and Petter Holme (Eds.), Springer, Singapore (2017), pp. 1–16.

2. Leo Speidel, Konstantin Klemm, Víctor M. Eguíluz, Naoki Masuda.
Epidemic threshold in temporally-switching networks.
In: Temporal Network Epidemiology, Naoki Masuda and Petter Holme (Eds.), Springer, Singapore (2017), pp. 161–177.

1. Naoki Masuda, Taro Takaguchi, Nobuo Sato, Kazuo Yano.
Self-exciting point process modeling of conversation event sequences.
In: Temporal Networks, P. Holme and J. Saramäki (Eds.), Springer-Verlag, Berlin (2013), pp. 245–264.

Refereed Conference Papers

11. Penghang Liu, Bahadir Altun, Rupam Acharyya, Robert E. Tillman, Shunya Kimura, Naoki Masuda, Ahmet Erdem Sarıyüce.
Temporal motifs for financial networks: A study on Mercari, JPMC, and Venmo platforms.
17th International Conference on Social Networks Analysis and Mining (ASONAM 2025).
Niagara Falls, Ontario, Canada, 8/25/2025–8/28/2025
In: Social Networks Analysis and Mining — 17th International Conference, ASONAM 2025, Niagara Falls, ON, Canada, August 25–28, 2025, Proceedings, Part I (Aijun An, Alfredo Cuzzocrea, Hongxin Xu (Editors)), Springer, Cham, Switzerland.
Lecture Notes in Computer Science, 16322, 211–226 (2026).
[Oral presentation; ≈ 25% acceptance]

10. Masaki Ogura, Junpei Tagawa, Naoki Masuda.
Distributed agreement on activity driven networks.
2018 American Control Conference.
Wisconsin Center, Milwaukee, Wisconsin, USA, 6/27/2018–6/29/2018
In: Proceedings, 4147–4152.
[Oral presentation]

9. Xia Cui, Sadamori Kojaku, Naoki Masuda, Danushka Bollegala.
Solving feature sparseness in text classification using core-periphery decomposition.
The Seventh Joint Conference on Lexical and Computational Semantics (*SEM 2018).
New Orleans, Lousiana, USA, 6/5/2018–6/6/2018
In: Proceedings, 255–264.
[Oral presentation; 42% acceptance]

8. Naoki Masuda, Konstantin Klemm, Víctor M. Eguíluz.
   Slowing down of linear consensus dynamics on temporal networks: some theoretical extensions.
   4th IFAC Conference on Analysis and Control of Chaotic Systems.
   Tokyo, Japan, 8/26/2015–8/28/2015
   In: IFAC-PapersOnLine, 48–18, 187–192 (2015).
   [Oral presentation]

7. Kodai Saito, Naoki Masuda.
   Two types of Twitter users with equally many followers.
   The 2013 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2013).
   Niagara Falls, Canada, 8/25/2013–8/28/2013
   In: Proceedings, 1425–1426.
   [Poster presentation; 36% acceptance (28% oral + 8% poster)]

6. Naoki Masuda, Tetsuya Fujie, Kazuo Murota.
   Application of semidefinite programming to maximize the spectral gap produced by node removal.
   4th Workshop on Complex Networks (CompleNet 2013).
   Berlin, Germany, 3/13/2013–3/15/2013
   In: Complex Networks IV, Studies in Computational Intelligence, 476, 155–163 (2013).
   [Poster presentation]

5. Naoki Masuda, Hiroshi Kori.
   STDP enhances frequency synchrony in neural networks with a pacemaker.
   International Joint Conference on Neural Networks (IJCNN 2007).
   Orlando, Florida, USA, 8/12/2007–8/17/2007
   In: Proceedings of International Joint Conference on Neural Networks, 96–101 (2007).
   [Poster presentation]

4. Naoki Masuda, Shun-ichi Amari.
   Modeling memory transfer and savings in cerebellar motor learning.
   Neural Information Processing Systems (NIPS) 2005.
   Vancouver, Canada, 12/5/2005–12/10/2005
   In: Advances in Neural Information Processing Systems (Eds. Y. Weiss, B. Scholkopf, J. Platt), 18, 859–866 (2006).
   [Poster presentation; 25% acceptance]

3. Naoki Masuda, Kazuyuki Aihara.
   Collective behavior of pulse-coupled FitzHugh-Nagumo neurons.
   The 8th International Conference on Neural Information Processing (ICONIP 2001).
   Shanghai, China, 11/14/2001–11/18/2001
   In: Proceedings of ICONIP2001, Vol. 2, 910–915.
   [Oral presentation]

2. Naoki Masuda, Kazuyuki Aihara.
   Cryptosystems based on space-discretization of chaotic maps.
   The IEEE International Symposium on Circuits and Systems, 2001 (ISCAS 2001).
   Sydney, Australia, 5/6/2001–5/9/2001
   In: Proceedings of ISCAS 2001, III, 321–324 (2001).
   [Oral presentation]

1. Naoki Masuda, Kazuyuki Aihara.
   Cryptosystem based on a finite-state baker's map and its security analysis.

1999 International Symposium on Nonlinear Theory and its Applications (NOLTA99).
Hilton Waikoloa Village, Waikoloa, Hawaii, USA, 11/28/1999–12/2/1999
In: Proceedings of NOLTA'99, 613–616.
[Oral presentation]

Institutional Journal Papers (refereed)

2. Aoi Naito, Naoki Masuda, Tatsuya Kameda.
Collective intelligence under a volatile task environment: A behavioral experiment using social networks and computer simulations.
Interdisciplinary Information Sciences, 30, 1–12 (2024).

1. Akihiro Fujihara, Yusuke Ide, Norio Konno, Naoki Masuda, Hiroyoshi Miwa, Masato Uchida.
Limit theorems for the average distance and the degree distribution of the threshold network model.
Interdisciplinary Information Sciences, 15, 361–366 (2010).

Media articles featuring my work and related activity

4. Vendor offering citations for purchase is latest bad actor in scholarly publishing.
ScienceInsider (online news from Science), 2/12/2024
Quoting my comments that are distantly related to Sadamori Kojaku, Giacomo Livan, Naoki Masuda.
Detecting citation cartels in journal networks.
Scientific Reports, 11, 14524 (2021).

3. Riconoscimento o gratitudine, le due ricompense dell'altruista.
le Scienze (in Italian), 3/5/2014
Featuring Takamitsu Watanabe, Masanori Takezawa, Yo Nakawake, Akira Kunimatsu, Hidenori Yamasue, Mitsuhiro Nakamura, Yasushi Miyashita, Naoki Masuda.
Two distinct neural mechanisms underlying indirect reciprocity.
Proceedings of the National Academy of Sciences of the United States of America, 111, 3990–3995 (2014).

2. Social networks, suicide and statistics.
I Programmer news, 7/6/2012
Featuring Naoki Masuda, Issei Kurahashi, Hiroko Onari.
Suicide ideation of individuals in online social networks.
PLoS ONE, 8, e62262 (2013).

1. Spotting suicidal tendencies on social networks.
MIT Technology Review, 7/5/2012
Featuring the same article.

Conference/workshop Invited Presentations (without refereed proceeding papers)

54. Naoki Masuda.
Effects of degree-heterogeneous networks on cooperation: Reconciling gaps between two major approaches.
"Evolutionary Games: Mathematical Theory and Biological Insights" workshop.
National Institute for Theory and Mathematics in Biology, Chicago, IL, 5/11/2026–5/15/2026
[This was an invitation-based workshop.]

53. Naoki Masuda.
Evolving evolutionary graph theory: multilayer networks, temporal networks, and eco-evolutionary consideration.
International Conference on "Cooperative Strategies and Dynamics in Social Systems: from Human to Insect Societies".
Meiji University, Tokyo, Japan, 1/14/2026–1/16/2026

52. Naoki Masuda.
Node-state dynamics view of temporal networks.
Satellite Workshop: TENET (TEmporal NETworks), NetSci 2025.
Maastricht, Netherlands, 6/2/2025–6/6/2025
[Keynote talk]

51. Naoki Masuda.
Early warning indicators for regime shifts in network dynamics.
Emergent Behavior in Complex Systems of Interacting Agents.
Institute for Mathematical and Statistical Innovation workshop, University of Chicago, Chicago, IL, 3/17/2025–3/21/2025
[This was an invitation-based workshop.]

50. Naoki Masuda.
Competition in dynamic populations and networks: skaters, pigeons, and cooperators.
Mathematics & Social Sciences Workshop
Institut d'Études Advancées, Paris, France, 2/24/2025–2/26/2025
[This was an invitation-based workshop.]

49. Naoki Masuda.
Fraud detection in network data: illegal online transactions and citation cartels.
The First Social Intelligence Conference (translation by NM).
Dalian, China, 9/13/2024
[Invited talk]

48. Naoki Masuda.
Network science, temporal networks, and interaction among manga characters.
(Workshop part of) the First Social Intelligence Conference and Workshop (translation by NM).
Dalian University of Technology, Dalian, China, 9/12/2024
[Invited talk]

47. Naoki Masuda.
Reinforcement learning and games in sports: Football coach's decision making and skater's dilemma.
2024 Evolutionary Game Theory and Artificial Intelligence International Conference (translation by NM).
Vision Hotel Beijing, Beijing, China, 8/5/2024–8/8/2024
[Plenary talk]

46. Naoki Masuda.
Recurrence view of temporal network data.
SIAM UB (University at Buffalo) Student Chapter Applied Mathematics Workshop.
Online, 4/26/2024
[Invited talk]

45. Naoki Masuda.
Correlation network analysis.

Session on Network Analysis and Subsampling Methods. UPSTAT 2024 — Statistical Science at a Crossroads.
Rochester Institute of Technology, Rochester, NY, 4/13/2024
[Invited talk]

44. Naoki Masuda.
Fixation dynamics on multilayer networks.
Workshop on Modeling and Applications of Evolutionary Game Theory.
Online, 12/8/2023
[Invited talk]

43. Naoki Masuda.
Fixation probability in opinion formation dynamics on higher-order networks.
The Future of Mathematical Social Science Workshop.
Online (hosted by University of Pennsylvania), Philadelphia, PA, 4/10 – 4/11/2023
[Invited talk]

42. Naoki Masuda.
Modeling and analysis of temporal networks and dynamical processes on them.
International Conference on Emerging Trends in Mathematical Sciences & Computing (IEMSC-23).
Online (hosted by IEM Kolkata, India), February 3–5, 2023.
[Plenary talk]

41. Naoki Masuda.
Long-tailed distributions of inter-event times as mixtures of exponentials.
KIAS International Workshop: Theoretical Challenges in Network Science.
KIAS, Seoul, South Korea, September 26–30, 2022.
[Invited talk]

40. Naoki Masuda.
State dynamics view of temporal network data: Recurrence and embedding.
Blockchain Kaigi 2022 (BCK22).
Tohoku University, Sendai, Japan, August 4–5, 2022.
[Invited talk]

39. Naoki Masuda.
Recurrence view of temporal network data: System-state dynamics, recurrence plot, and embedding.
NetSci 2022.
Online, July 25–29, 2022.
[Invited talk]

38. Naoki Masuda.
Recurrence quantification analysis and energy landscape analysis of dynamic brain networks. The BIRS workshop on Multiplex Brain Networks.
Banff International Research Station, Banff, Canada, 4/22/2022–4/24/2022.
[Invited talk]

37. Naoki Masuda.
System-state dynamics and recurrence of temporal networks.
American Physical Society March Meeting, Chicago, IL, USA, 3/14/2022–3/18/2022.
[Invited talk]

36. Naoki Masuda.
Network analysis on correlation matrices: Alternatives to threshold networks.
International Conference on AI and Data Science: Mathematics and Applications.
Suwon, South Korea, 11/4/2019–11/5/2019.
[Keynote talk]

35. Naoki Masuda.
Network science: Static and temporal networks.
2019 Big Data Analysis and Complex Science Management Summer School (translation by NM).
Harbin Institute of Technology (Weihai), Weihai, China, July 25–31, 2019.
[Invited lectures]

34. Naoki Masuda.
Network dynamics: Epidemic processes and energy landscape analysis.
2019 Annual Meeting, Society for Mathematical Biology.
Université de Montréal, Montreal, Canada, July 22–26, 2019.
[Plenary speaker]

33. Naoki Masuda.
Networks from correlation matrices: An alternative to thresholding. Threshold Networks. University of Nottingham, Nottingham, UK, July 22–24, 2019.
[Invited talk]

32. Naoki Masuda.
Lecture 1: Fundamentals of random walks on networks.
Lecture 2: Applications of random walks on networks.
CCSS (Center for Computational Social Science) School on Computational Social Science.
Kobe University, Japan, June 20–22, 2019.
[Invited lectures]

31. Naoki Masuda.
Contagion processes on temporal networks: Epidemic threshold, commutator, and concurrency.
Centre for Business Network Analysis (CBNA) Summer School in Social Network Analysis 2019.
University of Greenwich, London, UK, June 10–14, 2019.
[Invited talk]

30. Naoki Masuda.
A configuration model for correlation matrices.
4th Workshop on Statistical Physics for Financial and Economic Networks.
NetSci 2019, University of Vermont, Burlington, VA, USA, May 27–31, 2019
[Invited talk]

29. Naoki Masuda.
Opening Symposium of the Dutch Network Science Society.
TU Delft, Delft, Netherlands, 7 May, 2019.
[10 min pitch talk]

28. Naoki Masuda.
Epidemic processes on dynamically switching networks: Effects of commutator and concurrency.
Young European Probabilists (YEP) XV "Information Diffusion on Random Networks" Workshop.
TU Eindhoven, Eindhoven, Netherlands, March 25-29, 2019.
[Invited talk]

27. Naoki Masuda.
    Epidemic processes and random walks on time-varying graphs.
    Fundamentals of complex networks: From static towards evolving.
    University of Duisburg-Essen, Essen, Germany, March 6–8 (2019).
    [Invited mini lecture talks; 4 talks $\times$ 1 hr]

26. Naoki Masuda.
    Random walks on networks.
    Online Winter School on Spectral Methods for Complex Systems.
    Webinar, 1/21/2019–1/30/2019.
    [Video lecture presentation]

25. Naoki Masuda.
    Directed acyclic graphs in network science.
    International Workshop on Theoretical Perspectives in Network Science.
    Seoul National University, Seoul, South Korea, 12/7/2018–12/9/2018.
    [Invited talk]

24. Naoki Masuda.
    Network epidemiology.
    Summer Boot Camp of Infectious Disease Modeling, 2018.
    The Institute of Statistical Mathematics, Tachikawa, Japan, 8/3/2018.
    [Invited lecture]

23. Naoki Masuda.
    A Gillespie algorithm for simulating interacting non-Markovian point processes.
    2nd Symposium on Spatial Networks.
    University of Oxford, UK. September 13–14 (2017).
    [Invited talk]

22. Naoki Masuda.
    Network epidemiology.
    Summer Boot Camp of Infectious Disease Modeling, 2017.
    The Institute of Statistical Mathematics, Tachikawa, Japan, 8/3/2017
    [Invited lecture]

21. Naoki Masuda.
    Reinforcement learning behaviour by football managers.
    Push the Envelope of Statistical Physics: Econo, Social, Bio and Beyond.
    POSTECH, Pohang, Korea, 12/12/2016–12/13/2016.
    [Invited talk]

20. Naoki Masuda.
    Highly heterogeneous performances across human individuals: quantifications and implications.
    Push the Envelope of Statistical Physics: Econo, Social, Bio and Beyond.
    POSTECH, Pohang, South Korea, 12/12/2016–12/13/2016.
    [Ignite talk]

19. Naoki Masuda.
    Network analysis: An overview.
    Systems-NET/IDC in Systems Workshop: At the Intersect between Network Science and Systems Engineering.

University of Bristol, Bristol, UK, 9/12/2016.
[Invited talk]

18. Leo Speidel, Konstantin Klemm, Victor M. Eguíluz, Naoki Masuda.
Dynamical switching of networks facilitates endemicity in the susceptible-infected-susceptible epidemic model.
Asia-Pacific Econophysics Conference 2016.
University of Tokyo, Tokyo, August 24–26 (2016).
[Invited talk]

17. Naoki Masuda.
Cooperation under indirect reciprocity: evidence from neuroimaging.
The Third International Workshop on Market Design Technologies for Sustainable Development.
Keio University, Yokohama, Japan, August 21–22 (2015).
[Invited talk]

16. Naoki Masuda.
Opinion control in the voter model on networks.
Satellite Symposium: Information, Self-Organizing Dynamics and Synchronization (ISDOS) on Networks II. NetSci 2015.
Zaragoza, Spain, June 1–5 (2015).
[Invited talk]

15. Naoki Masuda.
Dynamics of opinion formation and cooperation in populations composed of heterogeneous agents.
The 3rd Sociophysics Workshop "From Opinion Dynamics to Voting, Conflict and Terrorism".
CEVIPOF, CNRS, Paris, 3/30/2015–3/31/2015.
[Invited talk]

14. Naoki Masuda.
Dominance hierarchy of worker ants as directed networks.
NetSci 2014.
Berkeley, CA, USA, June 2–6 (2014).
[Invited talk]

13. Naoki Masuda.
Laplacian dynamics slows down on temporal networks.
Network Frontier Workshop.
Northwestern University, Evanston, IL, USA, December 4–6 (2013).
[Invited talk]

12. Naoki Masuda, Konstantin Klemm, Víctor M. Eguíluz.
Laplacian dynamics on temporal networks.
International Workshop on Phase Transition, Critical Phenomena and Related Topics in Complex Networks.
Hokkaido University, Sapporo, Japan, September 9–11 (2013).
[Invited talk]

11. Naoki Masuda.
Laplacian-driven dynamics on temporal networks.
Workshop: Temporal and Dynamic Networks: From Data to Models. NetSci 2013.
Danish Technical University, Lyngby, Denmark, June 3–7 (2013).
[Invited talk]

10. Naoki Masuda.
    Temporal networks: Introduction.
    School. NetSci2013.
    Danish Technical University, Lyngby, Denmark, June 3–7 (2013).
    [Invited lecture]

9. Naoki Masuda.
   Quantifying importance of interaction events in temporal networks.
   Toward Mathematical Foundations of Complex Network Theory.
   Kyoto University, Kyoto, Japan, September 14–16 (2012).
   [Invited speaker]

8. Naoki Masuda.
   Laplacian-based centrality in directed graphs.
   Discrete Geometric Analysis.
   Kyoto University, Kyoto, Japan, August 27–31 (2012).
   [Invited talk]

7. Naoki Masuda.
   Dependence of oscillation regularity on structure of networks.
   Anomalous Statistics, Generalized Entropies and Information Geometry.
   Nara Women's University, Nara, Japan, March 6–10 (2012).
   [Invited talk]

6. Naoki Masuda, Jin Seop Kim, Byungnam Kahng.
   Priority queues with scale-free arrivals of incoming tasks.
   Applications of Physics in Financial Analysis 7th International Conference (APFA).
   Tokyo Institute of Technology, Tokyo, March 1–5 (2009).
   [Invited talk]

5. Naoki Masuda.
   How to set payoff matrices for evolutionary games on heterogeneous networks?
   3rd International Nonlinear Science Conference.
   Tokyo, Japan, March 13–15 (2008).
   [Invited talk]

4. Naoki Masuda.
   Evolutionary game dynamics on networks: cost normalization perspective.
   NSC Winter Workshop 2008: Complex Nonlinear Dynamics Ranging from Biology to Engineering.
   Hokkaido University, Sapporo, Japan, March 8–9 (2008).
   [Invited talk]

3. Naoki Masuda, Norio Konno.
   Multi-state interacting particle systems on scale-free networks.
   International Symposium on Topological Aspects of Critical Systems and Networks.
   Hokkaido University, Sapporo, Japan, February 13-14 (2006).
   In: Proceedings of the International Symposium on Topological Aspects of Critical Systems and Networks, 11–17.
   [Invited talk]

2. Naoki Masuda.
   Multi-state epidemic dynamics on complex networks.
   International Workshop on Complex Networks.

Seoul National University, Seoul, South Korea, June 23–24 (2005).
[Invited talk]

1. Naoki Masuda, Kazuyuki Aihara.
   Dependence of neural ergodicity on noise strength.
   International Symposium on Nonlinear Theory and its Applications (NOLTA2004).
   Fukuoka, Japan, November 29 – December 3 (2004).
   In: Proceedings of NOLTA2004, 35–38.
   [Invited talk]

Research Talks in Seminars (For internal talks, I only show formal seminar series talks.)

Midwest Mathematical Biology Seminar.
Online, 5/5/2026

Data-Enabled Sciences Seminar,
Department of Mathematics, University of Houston, Houston, TX, 3/6/2026

Complex Systems Seminar Series,
University of Michigan, Ann Arbor, MI, 2/3/2026

VT MathBio Seminar,
Virginia Tech, Blacksburg, VA, 10/8/2025

Illinois Mathematical Biology Seminar,
University of Illinois Urbana-Champaign, Champaign, IL, 4/2/2025

Applied Math Colloquium,
University of North Carolina, Chapel Hill, NC, 10/16/2024

[Delivered online] Colloquium, Graduate School of Culture Technology, Korea Advanced Institute of Science and Technology (KAIST), 9/10/2024

School of Economics and Management, Dalian University of Technology, Dalian, China, 9/6/2024

College of Engineering, Peking University, Beijing, China, 8/13/2024

Mathematical Institute, University of Oxford, Oxford, UK, 5/23/2024

Student Mathematics Colloquium, Niagara University, Niagara, NY, USA, 3/27/2024

Seminar, Santa Fe Institute, Santa Fe, NM, USA, 2/19/2024

CoCo Seminar, Binghamton University, Binghamton, NY, USA, 2/9/2024

Department of Mathematical Sciences Colloquium, New Mexico State University, Las Cruces, NM, USA, 12/1/2023

Seminar for Bioinformatics and Mathematics Colloquium, Florida State University, Tallahassee, FL, USA, 10/11/2023 and 10/13/2023, respectively

Applied Mathematics and Computation Seminar, University of Massachusetts Amherst, Amherst, MA, USA, 9/26/2023

College of Engineering, Peking University, Beijing, China, 6/16/2023

Tsinghua Laboratory of Brain and Intelligence Seminar Series, Tsinghua University, Beijing, China, 6/15/2023

University of Michigan, MI, USA, 2/8/2023

Graduate School of Medicine, Yonsei University, Seoul, South Korea, 9/28/2022

Center for Brain Science, RIKEN, Wako, Japan, 7/15/2022

Okinawa Institute of Science and Technology Graduate University, Onna, Japan, 7/6/2022

Clarkson Center for Complex Systems Science seminar, Clarkson University, Potsdam, NY, USA, 4/8/2022

Network Science Institute, Northeastern University, Boston, MA, USA, 12/3/2021

CoCo Seminar, Binghamton University, Binghamton, NY, USA, 9/29/2021

[Delivered online] Mathematical Biology Seminar, University of Pennsylvania, Philadelphia, PA, USA, 3/2/2021

[Delivered online] IMT School For Advanced Studies Lucca, Lucca, Italy, 1/22/2021

[Delivered online] WINPEC Seminar / Causal Inference and Data Science Research Group Seminar, Waseda University, Tokyo, Japan, 7/21/2020

Colloquium, Department of Physics and Astronomy, University of Rochester, Rochester, NY, USA, 11/13/2019.

Forum, Center for Brain Science, RIKEN, Wako, Japan, 7/12/2019

Graduate School of Information Sciences, Tohoku University, Sendai, Japan, 7/10/2019

Department of Public Health and Primary Care, Imperial College London, UK, 6/26/2019

Department of Intelligent Systems, TU Delft, Netherlands, 5/9/2019

Disease Ecology Seminar, Center for the Ecology of Infectious Diseases, University of Georgia, GA, USA, 3/20/2019

School of Natural Sciences and Humanities, Harbin Institute of Technology (Shenzhen), Shenzhen, China, 2/27/2019

School of Science, Harbin Institute of Technology (Shenzhen), Shenzhen, China, 2/26/2019

Department of Mathematics, Imperial College London, UK, 11/20/2018

Department of Computer Science, University of Exeter, UK, 10/25/2018

Institute for Cross-Disciplinary Physics and Complex Systems (IFISC), University of the Balearic Islands, Palma de Mallorca, Spain, 9/20/2018

Faculty of Management and Economics, Dalian University of Technology, Dalian, China, 9/5/2018

Graduate School of Informatics, Kyoto University, Kyoto, Japan, 7/20/2018

Institute of Industrial Science, University of Tokyo, Tokyo, Japan, 7/3/2018

School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, 6/28/2018

Faculty of Management and Economics, Dalian University of Technology, Dalian, China, 4/26/2018

LSI (Living Systems Institute) Seminar, University of Exeter, Exeter, UK, 3/20/2018

Network Science Institute, Northeastern University, Boston, MA, USA, 3/9/2018

School of Biological Sciences Research Seminar Series, University of Reading, UK, 1/23/2018

Centre for Water Systems, University of Exeter, UK, 10/20/2017

EPSRC Centre for Predictive Modelling in Healthcare, University of Exeter, UK, 4/3/2017

Department of Computer Science, University of Liverpool, UK, 3/14/2017

Department of Mathematics, University of Sussex, UK, 2/9/2017

Complex Systems Seminar, School of Mathematical Sciences, Queen Mary University of London, UK, 2/7/2017

MACSI seminar, Department of Mathematics and Statistics, University of Limerick, Ireland, 1/9/2017

Math Colloquium, Dartmouth College, USA, 11/17/2016

Networks and Collective Behaviour seminar series, University of Bath, UK, 2/18/2016

Networks Journal Club, Mathematical Institute, University of Oxford, UK, 2/4/2016

LAND Seminar, School of Mathematics, University of Leeds, UK, 6/23/2015

Cambridge Networks Network Seminar, University of Cambridge, UK, 12/2/2014

Centre for Neuroimaging Sciences, King's College London, UK, 9/19/2014

School of Computing Science, Newcastle University, UK, 5/29/2014

Department of Energy Science, Sungkyunkwan University, South Korea, 8/1/2012

Contributed (and Other) Conference/Workshop Oral Presentations (excluding those listed above)

156. Neil G. MacLaren, Baruch Barzel, Naoki Masuda.
Observing network dynamics by a small number of sentinel nodes.
NetSci 2026, Boston, MA, USA, 6/1/2026–6/5/2026

155. Naoki Masuda.
Multilayer gene network analysis: outliers, gene modules, and switch-like genes.
CeLiSIS Symposium: Frontiers of Transcriptomic Data Analysis: Single-Cell and Time-Series Inference of Cellular Dynamics and Regulatory Networks.
Kyoto University, Kyoto, Japan, 5/22/2026

154. Naoki Masuda.
Observation of network dynamics through sentinel nodes.
Minisymposium: Data-driven Learning and Reduced Order Modeling of Interacting Particle Systems.
SIAM Conference on Uncertainty Quantification (UQ26),
Minneapolis, MI, 3/22/2026

153. Chirantha Piyamal Bandara, Naoki Masuda.
Enhancing spatial early warning signals: Optimized node selection for tipping point detection.
14th International Conference on Complex Networks and their Applications.
Binghamton University, Vestal, NY, 12/9/2025–12/11/2025

152. Claudia Neuendorf, Naoki Masuda, Kou Murayama.
Classroom social network structure enhances prediction of educational outcomes: A large-scale study.
14th International Conference on Complex Networks and their Applications.
Binghamton University, Vestal, NY, 12/9/2025–12/11/2025

151. Naoki Masuda.
Anticipating regime shifts in biological complex systems using network analysis.
Special Symposium: Monitoring, Modelling and Predicting Biological Oscillations: From Genes to Ecosystems.
Joint Conference of Asian Conference on Mathematical Biology and Annual Meeting of Japanese

Society for Mathematical Biology.
Kyoto Terrsa, Kyoto, Japan, 7/7/2025–7/11/2025

150. Lucas Lacasa, F. Javier Marín-Rodríguez, Naoki Masuda, Lluís Arola-Fernández.
Scalar embedding of temporal network trajectories.
Dynamics Days Europe 2025.
Thessaloniki, Greece, 6/23/2025–6/27/2025

149. Ruodan Liu, Jnanajyoti Bhaumik, Naoki Masuda.
Fixation dynamics on various networks.
NetSci 2025
Maastricht, Netherlands, 6/2/2025–6/6/2025

148. Shilong Yu, Neil G. MacLaren, Naoki Masuda.
Using sample cross covariance to potentially improve early warning signals for network dynamics.
NetSci 2025
Maastricht, Netherlands, 6/2/2025–6/6/2025

147. Hiroki Sayama, Naoki Masuda.
Deriving dynamical model equations from temporal network data using a graph rewriting framework: A project update.
Satellite Workshop: TENET (TEmporal NETworks), NetSci 2025.
Maastricht, Netherlands, 6/2/2025–6/6/2025

146. Saiful Islam, Alber Aqil, Faraz Hach, Ibrahim Numanagić, Naoki Masuda, Omer Gokcumen.
Understanding segmental duplications and genomic evolvability through network analysis.
Great Lakes Bioinformatics Conference (GLBIO).
University of Minnesota, Minneapolis, MN, USA, 5/12/2025–5/15/2025

145. Naoki Masuda.
Eco-evolutionary game dynamics in a network with two communities.
SIAM Conference on Dynamical Systems.
Evolutionary Dynamics in Ecological, Biological and Social Systems minisymposium.
Denver, CO, 5/11/2025–5/15/2025

144. Chirantha Piyamal Tharusha Bandara, Naoki Masuda.
Enhancing spatial early warning signals: Optimized node selection for tipping point detection.
Eighth Northeast Regional Conference on Complex Systems.
Binghamton University, Vestal, NY, 4/9/2025–4/11/2025

143. Bisna Mary Eldo, Sarbendu Rakshit, Naoki Masuda.
Dimension reduction for graphon nonlinear dynamical systems.
Eighth Northeast Regional Conference on Complex Systems.
Binghamton University, Vestal, NY, 4/9/2025–4/11/2025

142. Saiful Islam, Alber Aqil, Omer Gokcumen, Naoki Masuda.
Interpretable community detection in multilayer gene co-expression networks.
Eighth Northeast Regional Conference on Complex Systems.
Binghamton University, Vestal, NY, 4/9/2025–4/11/2025

141. Neil Maclaren, Kazuyuki Aihara, Naoki Masuda.
Spatial early warning signals for dynamics on complex networks.
Eighth Northeast Regional Conference on Complex Systems.
Binghamton University, Vestal, NY, 4/9/2025–4/11/2025

140. Sarbendu Rakshit, Naoki Masuda.
Low-dimensional approximations in hypergraph dynamical systems.
Eighth Northeast Regional Conference on Complex Systems.
Binghamton University, Vestal, NY, 4/9/2025–4/11/2025
[Presented online]

139. Chanon Thongprayoon, Naoki Masuda.
Tie-decay networks with splines.
Eighth Northeast Regional Conference on Complex Systems.
Binghamton University, Vestal, NY, 4/9/2025–4/11/2025
[Best Oral Presentation Award (the first prize among 45 oral presentations)]

138. Shilong Yu, Neil MacLaren, Naoki Masuda.
Using cross-covariance of node states to potentially improve early warning signals.
Eighth Northeast Regional Conference on Complex Systems.
Binghamton University, Vestal, NY, 4/9/2025–4/11/2025

137. Jnanajyoti Bhaumik, Naoki Masuda.
Effects of edge weights on evolutionary dynamics.
NetSci-X, Indore, India, 1/15/2025–1/17/2025

136. Bisna Mary Eldo, Sarbendu Rakshit, Naoki Masuda.
Dimensional reduction of dynamical systems on graphons.
2024 SIAM New York-New Jersey-Pennsylvania Section Conference.
Rochester Institute of Technology, Rochester, NY, 11/1/2024–11/3/2024.

135. Saiful Islam, Naoki Masuda.
Detecting interpretable communities in multilayer correlation networks.
2024 SIAM New York-New Jersey-Pennsylvania Section Conference.
Rochester Institute of Technology, Rochester, NY, 11/1/2024–11/3/2024.

134. Sarbendu Rakshit, Naoki Masuda.
Dimensional reduction for hypergraph dynamical systems.
2024 SIAM New York-New Jersey-Pennsylvania Section Conference.
Rochester Institute of Technology, Rochester, NY, 11/1/2024–11/3/2024.

133. Maisha Islam Sejunti, Dane Taylor, Naoki Masuda.
Parrondo's paradox in the dynamics of the susceptible-infectious-susceptible epidemic model over periodic temporal networks.
2024 SIAM New York-New Jersey-Pennsylvania Section Conference.
Rochester Institute of Technology, Rochester, NY, 11/1/2024–11/3/2024.

132. Shilong Yu, Naoki Masuda.
Using off-diagonal entries of the covariance matrix to determine an optimized node set for early warning signals.
2024 SIAM New York-New Jersey-Pennsylvania Section Conference.
Rochester Institute of Technology, Rochester, NY, 11/1/2024–11/3/2024.

131. Hang-Hyun Jo, Tibebe Birhanu, Naoki Masuda.
Temporal scaling theory for bursty time series with clusters of arbitrarily many events.
2024 Korean Physical Society Fall Meeting.
Yeosu Expo Convention Center, Yeosu, South Korea, 10/22/2024–10/26/2024

130. Jnanajyoti Bhaumik, Naoki Masuda.
Constant selection on weighted networks.
American Mathematical Society 2024 Fall Eastern Sectional Meeting.
University at Albany, Albany, NY, 10/19/2024–10/20/2024

129. Naoki Masuda.
Modeling of animal collective behavior: many-wrongs hypothesis for pigeons and social brain hypothesis for primates. Special Session on Agent-Based and Mean-Field Modeling for Complex Social Systems.
American Mathematical Society 2024 Fall Eastern Sectional Meeting.
University at Albany, Albany, NY, 10/19/2024–10/20/2024

128. Neil G. MacLaren, Baruch Barzel, Naoki Masuda.
Approximating dynamics on networks: Hub nodes make poor sentinels.
NetSci 2024, Quebec City, Canada, 6/16/2024–6/21/2024

127. Sarbendu Rakshit, Naoki Masuda.
Dimensional reduction for hypergraph dynamical systems.
NetSci 2024, Quebec City, Canada, 6/16/2024–6/21/2024

126. Hiroki Sayama, Naoki Masuda.
Deriving dynamical model equations from temporal network data using a graph rewriting framework.
NetSci 2024, Quebec City, Canada, 6/16/2024–6/21/2024

125. Kashin Sugishita, Naoki Masuda.
Social network analysis of Japanese manga.
NetSci 2024, Quebec City, Canada, 6/16/2024–6/21/2024

124. Neil MacLaren, Prosenjit Kundu, Naoki Masuda.
Early warning signals for dynamics on networks.
SIAM Conference on the Life Sciences.
Portland, OR, 6/10/2024–6/13/2024

123. Saiful Islam, Naoki Masuda.
Detection of interpretable communities in multilayer biological networks using a penalized sparse factor model.
Session on Network Analysis and Subsampling Methods. UPSTAT 2024 — Statistical Science at a Crossroads.
Rochester Institute of Technology, Rochester, New York, 4/13/2024

122. Naoki Masuda.
Early warning signals for dynamics on networks.
Special session on "Complex Systems in the Life Sciences" at the American Mathematical Society (AMS) 2024 Spring Eastern Sectional Meeting.
Howard University, Washington DC, 4/6/2024–4/7/2024

121. Katherine Betz, Feng Fu, Naoki Masuda.
Group-structured evolutionary game dynamics with environmental feedback.
NERCCS 2024: Seventh Northeast Regional Conference on Complex Systems.
Clarkson University, Potsdam, NY, 3/20/2024–3/22/2024

120. Si Thu Aung, Mintao Liu, Uttara Tipnis, Kauser Abbas, Joaquín Goñi, Naoki Masuda.
Evaluating trajectories derived from dynamic functional connectivity across fMRI conditions.

NERCCS 2024: Seventh Northeast Regional Conference on Complex Systems.
Clarkson University, Potsdam, NY, 3/20/2024–3/22/2024

119. Jnanajyoti Bhaumik, Naoki Masuda.
Evolutionary dynamics on switching networks.
SIAM-NNP (New York-New Jersey-Pennsylvania) Conference.
New Jersey Institute of Technology, Newark, NJ, 10/20/2023–10/22/2023

118. Chanon Thongprayoon, Naoki Masuda.
Online and offline network embedding using landmarks.
SIAM-NNP (New York-New Jersey-Pennsylvania) Conference.
New Jersey Institute of Technology, Newark, NJ, 10/20/2023–10/22/2023

117. Hiroki Sayama, Naoki Masuda.
Deriving dynamical model equations from temporal network data using a graph rewriting framework.
Satellite session: Recent Advances in Learning and Data-driven Modeling of Complex Systems.
Conference on Complex Systems 2023.
Salvador, Brazil, 10/16/2023–10/20/2023

116. Naoki Masuda.
Epidemic threshold of a susceptible-infectious-susceptible model when node interaction obeys non-Poissonian statistics.
Special Session on Difference and Differential Equations: Modeling, Analysis, and Applications to Mathematical Biology.
American Mathematical Society, 2023 Fall Eastern Sectional Meeting.
State University of New York at Buffalo, Buffalo, NY, 9/9/2023–9/10/2023

115. Elohim Fonseca dos Reis, Naoki Masuda.
Metapopulation network models explain non-Poissonian statistics of intercontact times.
10th International Congress on Industrial and Applied Mathematics (ICIAM).
Waseda University, Tokyo, Japan, 8/20/2023–8/25/2023

114. Naoki Masuda.
Early warning signals for multistage transitions in tipping dynamics on networks.
Minisymposium: DNB Theory and its Applications, 10th International Congress on Industrial and Applied Mathematics (ICIAM).
Waseda University, Tokyo, Japan, 8/20/2023–8/25/2023

113. Naoki Masuda, Prosenjit Kundu.
Dimension reduction of dynamical systems on networks using various eigenvectors of adjacency matrices.
SIAM Conference on Applications of Dynamical Systems.
Portland, OR, 5/14/2023–5/18/2023

112. Naoki Masuda, Prosenjit Kundu.
Reducing the dimension of dynamical systems on networks using non-leading eigenvectors of adjacency matrices.
CompleNet 2023.
Universidade de Aveiro, Aveiro, Portugal, 4/25/2023–4/28/2023

111. Yiding Cao, Suraj Rajendran, Prathic Sundararajan, Royal Law, Sarah Bacon, Steven Sumner, Naoki Masuda.
Twitter following networks of individuals with adverse childhood experiences.

NERCCS 2023: Sixth Northeast Regional Conference on Complex Systems.
Clarkson University, Potsdam, NY, 3/22/2023–3/24/2023

110. Ruodan Liu, Naoki Masuda.
Fixation probability on hypergraphs.
NERCCS 2023: Sixth Northeast Regional Conference on Complex Systems.
Clarkson University, Potsdam, NY, 3/22/2023–3/24/2023

109. Neil MacLaren, Baruch Barzel, Naoki Masuda.
Low-dimensional approximations to nonlinear dynamics on networks: A sentinel node approach.
NERCCS 2023: Sixth Northeast Regional Conference on Complex Systems.
Clarkson University, Potsdam, NY, 3/22/2023–3/24/2023

108. Jnanajyoti Bhaumik, Naoki Masuda.
Fixation probability of switching networks.
11th International Conference on Complex Networks and Their Applications.
Palermo, Italy, 11/8/2022–11/10/2022

107. Prosenjit Kundu, Hiroshi Kori, Naoki Masuda.
Validity of a one-dimensional reduction of dynamical systems on networks.
11th International Conference on Complex Networks and Their Applications.
Palermo, Italy, 11/8/2022–11/10/2022

106. Neil Maclaren, Prosenjit Kundu, Naoki Masuda.
Early warning signals of multistage state transitions on complex networks.
11th International Conference on Complex Networks and Their Applications.
Palermo, Italy, 11/8/2022–11/10/2022

105. Kazuki Nakajima, Kazuyuki Shudo, Naoki Masuda.
Random hypergraph models preserving degree correlation and local clustering.
11th International Conference on Complex Networks and Their Applications.
Palermo, Italy, 11/8/2022–11/10/2022

104. Maisha Islam Sejunti, Naoki Masuda, Dane Taylor.
Floquet theory for spreading dynamics over periodically switching networks.
11th International Conference on Complex Networks and Their Applications.
Palermo, Italy, 11/8/022–11/10/2022

103. Qiong Zhang, Noha Abdel-Mottaleb, Kashin Sugishita, Naoki Masuda.
Bio-inspired design to enhance water distribution network performance.
Symposium Honoring John Crittenden at ACS Fall National Meeting, George Pullman (Marriott Marquis Chicago), Chicago, IL, 8/21/2022–8/25/2022

102. Elohim Fonseca dos Reis, Naoki Masuda.
Emergent non-Poissonian statistics of interevent times from metapopulation models.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, NY, 3/30/2022–4/1/2022

101. Ruodan Liu, Masaki Ogura, Elohim Fonseca Dos Reis, Naoki Masuda.
Impacts of concurrency on epidemic spreading in Markovian temporal networks.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, NY, 3/30/2022–4/1/2022

100. Neil Maclaren, Siobhán Mattison, Naoki Masuda.
A maximum entropy approach to the multivariate "space" of social networks.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, NY, 3/30/2022–4/1/2022
[Honorable Mention of the Best Oral Presentation Award]

99. Kazuki Nakajima, Kazuyuki Shudo, Naoki Masuda.
Higher-order rich-club phenomenon in research funding.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, NY, 3/30/2022–4/1/2022

98. Maisha Islam Sejunti, Naoki Masuda, Dane Taylor.
Floquet theory for spreading dynamics over periodically switching networks.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, NY, 3/30/2022–4/1/2022

97. Naoki Masuda.
A state-dynamics view of bursts in contact networks, with applications to epidemic processes and metapopulation models.
Special Session for Mathematical Methods for Ecology and Evolution in Structured Populations, Spring Eastern Virtual Sectional Meeting.
American Mathematical Society Spring Eastern Virtual Sectional Meeting. Online. 3/19/2022–3/20/2022

96. Ruodan Liu, Masaki Ogura, Elohim Fonseca dos Reis, Naoki Masuda.
Modeling effects of concurrency on epidemic spreading in Markovian temporal networks.
Networks 2021: A Joint Sunbelt and NetSci Conference.
Online, 7/5/2021–7/10/2021

95. Chanon Thongprayoon, Lorenzo Livi, Naoki Masuda.
Continuous-time trajectory of tie-decay temporal networks.
Networks 2021: A Joint Sunbelt and NetSci Conference.
Online, 7/5/2021–7/10/2021

94. Noha Abdel-Mottaleb, Qiong Zhang, Kashin Sugishita, Naoki Masuda.
Can bio-inspired network design improve the performance of water distribution networks (WDNs)?
2021 World Environmental & Water Resources Congress.
Online, 6/7/2021–6/11/2021

93. Alessio Cardillo, Naoki Masuda.
Tipping point in evolutionary games on networks triggered by zealots.
CompleNet Live 2021, Online, 5/24/2021–5/26/2021

92. Irene Malvestio, Alessio Cardillo, Naoki Masuda.
Interplay between k-core and communities in networks.
CompleNet Live 2021, Online, 5/24/2021–5/26/2021

91. Naoki Masuda, Victor M. Preciado, Masaki Ogura.
Analysis of the susceptible-infected-susceptible model in finite networks using the non-backtracking matrix.
Minisymposium on Latest Advances in Spectral Linear Algebra in Network Science, the SIAM Conference on Applied Linear Algebra (LA21).
Online, 5/17/2021–5/21/2021

90. Hang-Hyun Jo, Naoki Masuda.
Finite-size effects on the convergence time in continuous-opinion dynamics.
2021 Korean Physical Society Spring Meeting. Online, 4/21/2021–4/23/2021

89. Lingqi Meng, Naoki Masuda.
Epidemic threshold for metapopulation model networks with a second-order mobility rule.
NERCCS 2021: Fourth Northeast Regional Conference on Complex Systems.
Online, 3/31/2021–4/2/2021

88. Mengyuan J. Sun, Naoki Masuda.
COVID-19 impacts on domestic airlines via multilayer network analysis.
NERCCS 2021: Fourth Northeast Regional Conference on Complex Systems.
Online, 3/31/2021–4/2/2021

87. Chanon Thongprayoon, Lorenzo Livi, Naoki Masuda.
Visualizing trajectory of tie-decay temporal networks.
NERCCS 2021: Fourth Northeast Regional Conference on Complex Systems.
Online, 3/31/2021–4/2/2021

86. Lingqi Meng, Naoki Masuda.
Diffusion speed of the node2vec random walk on networks
COMPLEX20 (Conference on Complex Systems 2020).
Online, 12/4/2020–12/11/2020

85. Xiuxiu Zhan, Ziyu Li, Naoki Masuda, Petter Holme, Huijuan Wang.
Susceptible-infected-spreading-based network embedding in static and temporal networks.
9th International Conference on Complex Networks and their Applications.
Online, 12/1/2020–12/3/2020

84. Sadamori Kojaku, Giacomo Livan, Naoki Masuda.
Detecting citation cartels in journal citation networks.
NetSci 2020.
Online (due to COVID-19), 9/17/2020–9/25/2020

83. Irene Malvestio, Alessio Cardillo, Naoki Masuda.
Interplay between k-core and community structure in networks.
NetSci 2020.
Online (due to COVID-19), 9/17/2020–9/25/2020

82. Kashin Sugishita, Naoki Masuda.
Structural evolution of US airline networks.
NetSci 2020.
Online (due to COVID-19), 9/17/2020–9/25/2020

81. Hang-Hyun Jo, Takayuki Hiraoka, Naoki Masuda, Aming Li.
Modeling temporal networks with bursty activity patterns of nodes and links.
2020 Korean Physical Society Spring Meeting.
Online, 7/13/2020–7/15/2020

80. Marinho A. Lopes, Jiaxiang Zhang, Dominik Krzemiński, Khalid Hamandi, Qi Chen, Lorenzo Livi, Naoki Masuda.
Recurrence analysis of dynamic functional brain networks.
Third Northeast Regional Conference on Complex Systems.
Online, 4/1/2020–4/3/2020

79. Elohim Fonseca dos Reis, Aming Li, Naoki Masuda.
Interacting human dynamics as a mixture of Poisson processes.
Third Northeast Regional Conference on Complex Systems.
Online, 4/1/2020–4/3/2020

78. Alessio Cardillo, Naoki Masuda.
Critical mass effect in evolutionary games on networks triggered by zealots.
NetSci-X 2020.
Tokyo, Japan, 1/20/2020–1/23/2020

77. Kashin Sugishita, Mason A. Porter, Mariano Beguerisse-Díaz, Naoki Masuda.
Opinion dynamics in tie-decay networks.
NetSci-X 2020.
Tokyo, Japan, 1/20/2020–1/23/2020

76. Marinho A. Lopes, Jiaxiang Zhang, Dominik Krzemiński, Khalid Hamandi, Lorenzo Livi, Naoki Masuda.
Recurrence analysis of dynamic brain networks: Characterisation of the spatio-temporal dynamics of magnetoencephalographic recordings.
8th International Conference on Complex Networks and their Applications.
Lisbon, Portugal, 12/10/2019–12/12/2019

75. Elohim Fonseca dos Reis, Mark Viney, Naoki Masuda.
Network structure of wild and laboratory mice immune states.
International Conference on AI and Data Science: Mathematics and Applications.
Suwon, South Korea, 11/4/2019–11/5/2019

74. J. Miyata, A. Sasamoto, T. Ezaki, N. Masuda, Y. Mori, M. Isobe, T. Aso, T. Murai, H. Takahashi.
Conservative and hasty decision styles are differently associated with static and dynamic functional connectivity in healthy and schizophrenia people.
Society for Neuroscience, Chicago, IL, 10/19/2019–10/23/2019

73. Aoi Naito, Naoki Masuda, Tatsuya Kameda.
Social network and collective intelligence under nonstationary uncertain environment.
7th International Conference on Cognitive Neurodynamics.
Alghero, Italy, 9/29/2019–10/2/2019.

72. Takayuki Hiraoka, Naoki Masuda, Aming Li, Hang-Hyun Jo.
Modeling temporal networks with bursty node and link activity.
Conference on Complex Systems 2019.
Nanyang Technological University, Singapore, 9/30/2019–10/4/2019.

71. Elohim Fonseca dos Reis, Naoki Masuda, Mark Viney.
Immune networks in wild mice, *Mus musculus domesticus*.
Parasitic Helminths: New Perspectives in Biology and Infection.
Hydra, Greece, September 1–6, 2019.

70. Sadamori Kojaku, Giulio Cimini, Guido Caldarelli, Naoki Masuda.
Structural changes in an interbank market across the financial crisis: A multiple core-periphery analysis.
NetSci 2019, University of Vermont, Burlington, VA, USA, May 27–31, 2019.

69. Jun Miyata, Akihiko Sasamoto, Takahiro Ezaki, Naoki Masuda, Yasuo Mori, Masanori Isobe, Toshihiko Aso, Toshiya Murai, Hidehiko Takahashi.
Is jumping to conclusions bias associated with frequent “jumping” to salience-related functional

brain states?
2019 Congress of the Schizophrenia International Research Society, Orlando, Florida, USA, April 10–14, 2019.

68. Elohim Fonseca dos Reis, Mark Viney, Naoki Masuda.
Immune state networks of wild and laboratory mice.
10th International Conference on Complex Networks (COMPLNET'19).
Tarragona, Spain, March 18–21, 2019.

67. Naoki Masuda.
Energy landscape analysis of multivariate time series data.
New Trends in Statistical Physics.
Lucca, Italy, March 15 (2019).

66. Sadamori Kojaku, Mengqiao Xu, Haoxiang Xia, Naoki Masuda.
Multiscale core-periphery structure in a global liner shipping network.
The 7th International Conference on Complex Networks and Their Applications.
University of Cambridge, Cambridge, UK, December 11–13 (2018).

65. Takamitsu Watanabe, Geraint E. Rees, Naoki Masuda.
Neural retainability in autism.
Society for Neuroscience 2018.
San Diego, CA, November 3–7 (2018).

64. Naoki Masuda.
Energy landscape modelling of network dynamics.
GW4 Biomed MRC DTP, Interdisciplinary Advanced Training Workshop, Computational Modelling.
University of Bristol, Bristol, UK, 10/17/2018.

63. Takahiro Ezaki, Michiko Sakaki, Takamitsu Watanabe, Naoki Masuda.
Energy landscape analysis of age-related changes in human brain activity.
Conference on Complex Systems (CCS 2018).
Thessaloniki, Greece, September 23–28 (2018).

62. Genki Ichinose, Naoki Masuda.
Zero-determinant strategies in repeated prisoner's dilemma games.
ALIFE.
Tokyo, Japan, 7/23/2018–7/27/2018.

61. Tomokatsu Onaga, James P. Gleeson, Naoki Masuda.
Concurrency-induced transitions in epidemic processes on temporal networks.
NetSci 2018.
Paris, France, June 11–15 (2018).

60. Naoki Masuda.
Reinforcement learning explains conditional cooperation and its moody cousin in games on networks.
Sattelite meeting: GAMENET: Games on Networks, NetSci 2018.
Paris, France, June 11–15 (2018).

59. Naoki Masuda.
Detecting sequences of system states in temporal networks.
Symposium on Networks, Time and Causality.
ETH, Zurich, Switzerland, April 13 (2018).

58. Anne-Lene Sax, Liz Coulthard, Naoki Masuda.
Epidemic dynamics of disease progression in dementia – A numerical study.
Brain Networks and Neurological Disorders: from Theory to Clinic.
University of Exeter, Exeter, UK, April 10 (2018).

57. Naoki Masuda, Michiko Sakaki, Takahiro Ezaki, Takamitsu Watanabe.
Clustering coefficients for correlation matrices and their application to neuroimaging data.
NetSci-X.
Hangzhou, China, Jan 5–8 (2018).

56. Sadamori Kojaku, Naoki Masuda.
Core-periphery structure in degree-heterogeneous networks.
NetSci-X.
Hangzhou, China, Jan 5–8 (2018).

55. Naoki Masuda.
How do people sense population density to make migration decisions?
Maths and the City Workshop.
University of Bristol, Bristol, UK, January 19 (2017).

54. Sadamori Kojaku, Naoki Masuda.
Finding multiple core/periphery structure with random walks.
The 5th International Workshop on Complex Networks and Their Applications.
Milan, Italy, 11/30/2016–12/2/2016.

53. Kohei Tamura, Naoki Masuda.
Effects of spatial interaction on migration flows.
Asia-Pacific Econophysics Conference 2016.
University of Tokyo, Tokyo, Japan, August 24–26 (2016).

52. Teruyoshi Kobayashi, Naoki Masuda.
A community-based collective influence algorithm for immunizing networks.
Asia-Pacific Econophysics Conference 2016.
University of Tokyo, Tokyo, Japan, August 24–26 (2016).

51. Naoki Masuda, Luis E. C. Rocha.
Exact simulation of interacting non-Markovian renewal processes using the Laplace transform.
NetSci 2016.
Seoul, South Korea, 5/30/2016–6/3/2016.

50. Naoki Masuda.
Immunizing networks by targeting collective influencers at a community level.
Satellite workshop: When complex networks meet complex data: Higher-order models in network science. NetSci 2016.
Seoul, South Korea, 5/30/2016–6/3/2016.

49. Naoki Masuda.
Ranking professional tennis players using a temporal network centrality measure.
Satellite workshop: Competition networks and centrality, NetSci 2016.
Seoul, South Korea, 5/30/2016–6/3/2016.

48. Naoki Masuda.
Reinforcement learning explains conditional cooperation and its moody cousin.

GW4 Game Theory Workshop: Incentives for Conflict and Cooperation.
University of Bristol, Bristol, UK, May 9–10 (2016).

47. Naoki Masuda.
Individual-based approach to the susceptible-infected-recovered epidemic processes on arbitrary dynamic networks.
Dynamical Networks and Network Dynamics.
International Centre for Mathematical Sciences, Edinburgh, UK, January 18–22 (2016).

46. Luis E. C. Rocha, Naoki Masuda.
Individual-based approximation to the susceptible-infected-recovered model on temporal networks.
NetSci-X 2016.
Wroclaw, Poland, January 11–13 (2016).

45. Naoki Masuda, Konstantin Klemm, Víctor M. Eguíluz.
Laplacian dynamics on temporally switching networks.
IEEE International Meeting on Analysis and Applications of Nonsmooth Systems (AANS2014).
Como, Italy, September 10–12 (2014).

44. Hiroyuki Shimoji, Masato S. Abe, Kazuki Tsuji, Naoki Masuda.
Dominance hierarchy networks of ant workers.
The Joint Annual Meeting of the Japanese Society for Mathematical Biology and the Society for Mathematical Biology.
Osaka, Japan, July 28 – August 1 (2014).

43. Naoki Masuda.
Neural underpinning of cooperation under indirect reciprocity.
Making of Humanities: Biological Roots of Mathematics and Cooperation.
Hokkaido University, Sapporo, Japan, July 28 (2014).

42. Naoki Masuda.
Analysis of diffusive processes on temporal networks.
The 3rd International Symposium on Innovative Mathematical Modelling.
University of Tokyo, Tokyo, Japan, November 12–15 (2013).

41. Naoki Masuda.
Indirect reciprocity and ingroup favoritism: mathematical models.
International Workshop on Social Computing 2013 – Perspectives in Socio-Cultural Complexity.
Graduate School's Seoul Office, Seoul, South Korea, July 27 (2013).

40. Naoki Masuda.
Social spike trains: where computational neuroscience and computational social science can meet.
Modeling Neural Activity: Statistics, Dynamical Systems, and Networks.
Lihue, Hawaii, June 26–28 (2013).

39. Naoki Masuda.
Suicide ideation in online social networks.
Workshop: NetSciEd2. NetSci2013.
Danish Technical University, Lyngby, Denmark, June 3–7 (2013).

38. Mitsuhiro Nakamura, Naoki Masuda.
Sharing information in groups yields ingroup favoritism in indirect reciprocity.
Modelling Biological Evolution 2013: Recent Progress, Current Challenges and Future Directions.
University of Leicester, Leicester, UK, May 1–3 (2013).

37. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Network structure dependence of collective enhancement of temporal precision in coupled noisy oscillators.
The First Annual Winter q-bio Meeting.
Waikiki, Hawaii, USA, February 18–21 (2013).

36. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Dependence of oscillation regularity on network structure in coupled noisy oscillators.
European Conference on Complex Systems (ECCS'12).
Brussel, Belgium, September 3–7 (2012).

35. Taro Takaguchi, Nobuo Sato, Kazuo Yano, Naoki Masuda.
A centrality measure for interaction events in temporal networks.
European Conference on Complex Systems (ECCS'12).
Brussel, Belgium, September 3–7 (2012).

34. Taro Takaguchi, Naoki Masuda, Petter Holme.
Effect of bursty communication patterns on contagion in a threshold-based epidemic dynamics.
European Conference on Complex Systems (ECCS'12). Satellite meeting: Data-driven modeling of contagion processes.
Brussel, Belgium, September 3–7 (2012).

33. Taro Takaguchi, Nobuo Sato, Kazuo Yano, Naoki Masuda.
Detecting important interaction events in temporal networks.
NetSci 2012.
Northwestern University, Evanston, Illinois, USA, June 20–22 (2012).

32. Shoma Tanabe, Hideyuki Suzuki, Naoki Masuda.
Indirect reciprocity with three reputation values.
The 2nd International Symposium on Innovative Mathematical Modelling.
University of Tokyo, Tokyo, Japan, May 14–19 (2012).

31. Ryo Fujie, Kazuyuki Aihara, Naoki Masuda.
Conditions for stable coexistence of multiple species in consensus dynamics.
The 2nd International Symposium on Innovative Mathematical Modelling.
University of Tokyo, Tokyo, Japan, May 14–19 (2012).

30. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Network structure dependence of oscillation regularity in coupled noisy oscillators.
Engineering of Chemical Complexity.
Berlin, Germany, July 4–8 (2011).

29. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Design principle of precise neural clocks.
The 3rd International Conference on Cognitive Neurodynamics (ICCN 2011).
Hilton Niseko Village, Hokkaido, Japan, June 9–13 (2011).

28. Takamitsu Watanabe, Naoki Masuda.
Sequential node removal to increase the spectral gap of networks.
NetSci 2011.
Budapest, Hungary, June 8–10 (2011).

27. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Collective enhancement of temporal precision in networks of noisy oscillators.

SIAM Conference on Applications of Dynamical Systems.
Snowbird, Utah, USA, May 22–26 (2011).

26. Naoki Masuda, Yoji Kawamura, Hiroshi Kori.
Collective phase diffusion in networks of noisy oscillators.
SIAM Conference on Applications of Dynamical Systems. Workshop: Linking Neuronal Network Architecture and Collective Dynamics.
Snowbird, Utah, USA, May 22–26 (2011).

25. Naoki Masuda.
Laplacian-based centrality in directed networks.
Applications of Network Theory: From Mechanisms to Large-scale Structure.
Nordita, Stockholm, Sweden, April 18 (2011).

24. Hiroshi Kori, Yoji Kawamura, T. Okano, Naoki Masuda.
Collective phase diffusion and temporal precision in networks of noisy oscillators.
Dynamics Days Europe.
Bristol, UK, September 6–10 (2010).

23. Naoki Masuda.
Threshold networks with random weights and their extension to spatial networks.
The 34th Conference on Stochastic Processes and their Applications. Invited special session "Spatial Random Networks" organized by Remco van der Hofstad.
Osaka, Japan, September 6–10 (2010).

22. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Theory of collective enhancement of temporal precision in oscillator networks.
Biosignal 2010.
Berlin, Germany, July 14–16 (2010).

21. Takehisa Hasegawa, Naoki Masuda.
Information cascades with aging of nodes in networks.
The 9th Asia-Pacific Complex Systems Conference (Complex'09).
Tokyo, Japan, November 4–7 (2009).

20. Ralf Toenjes, Hiroshi Kori, Naoki Masuda.
Synchronization transition in sparse unidirectional networks of identical oscillators.
Frontiers in Network Science Advances and Applications.
Berlin, Germany, 9/28/2009–9/30/2009.

19. Jin Seop Kim, Naoki Masuda, Byungnam Kahng.
A priority queue model of human dynamics with bursty input tasks.
First International Conference on Complex Sciences: Theory and Applications (Complex'2009).
Shanghai, China, February 23–25 (2009).
In: Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, 5, 2402–2410 (2009).

18. Naoki Masuda, Hisashi Ohtsuki.
Evolutionary dynamics and fixation probabilities on directed and weighted networks.
International Conference on Complex Networks: The Past 10 Years and Future.
Seoul National University, Seoul, South Korea, December 19–22 (2008).

17. Naoki Masuda.
Dual coding by spiking neural networks.

NIPS workshop: Statistical analysis and modeling of response dependencies in neural populations, coordinated by A. Onken, K. Obermayer, V. Dragoi, S. Grunewalder, D. Berger.
Whistler, Canada, December 12–13 (2008).

16. Naoki Masuda, Hiroshi Kori.
Emergence of feedforward networks and entrainment in oscillator networks via a biological synaptic plasticity rule.
SigmaPhi – International Conference in Statistical Physics.
Chania, Greece, July 14–18 (2008).

15. Naoki Masuda, Taro Ueno.
Controls of nosocomial infection based on observed social networks of a community hospital.
NetSCI2008.
Norwich, UK, June 23–27 (2008).

14. S. Kondo, Naoki Masuda, H. Tanaka, Yuki Yasuda.
Network analysis of WWW with a website categorization database.
Fourth Joint Japan-North America Mathematical Sociology Conference.
Redondo Beach, CA, USA, May 29 – June 1 (2008).

13. Naoki Masuda.
Evolutionary games with participation costs on networks.
NetSci: International Workshop and Conference on Network Science.
New York, NY, USA, May 20–25 (2007).

12. Naoki Masuda, Norio Konno.
Cyclic competitive dynamics on networks.
Japanese-Korean Joint Meeting for Mathematical Biology.
Fukuoka, Japan, September 16–18 (2006).

11. Naoki Masuda, Norio Konno.
Rock-scissors-paper game on complex networks.
NetSci: International Workshop and Conference on Network Science.
Indiana University, Bloomington, Indiana, USA, May 22–25 (2006).

10. Yuichi Katori, Naoki Masuda, Kazuyuki Aihara.
Dynamic switching of neural coding schemes in a network with gap junctions.
Artificial Life and robotics (AROB) 2006.
Beppu, Japan, 1/23/2006–1/25/2006.

9. Naoki Masuda, Norio Konno.
Properties of complex networks generated with vertex fitness and homophily.
NEXT – Sigma Phi, News, Expectations and Trends in Statistical Physics.
Kolymbari, Greece, August 13–18 (2005).

8. Hiroyoshi Miwa, Naoki Masuda, Norio Konno.
Scale-free networks and network mining problems.
IFORS Triennial.
Honolulu, Hawaii, USA, July 11–15 (2005).

7. Naoki Masuda, Hiroyoshi Miwa, Norio Konno.
On a nongrowing small-world and scale-free network model with geographical consideration.
Complexity and Nonextensivity – New Trends in Statistical Mechanics.
Kyoto University, Kyoto, Japan, March 14–18 (2005).

6. Naoki Masuda, Kazuyuki Aihara.
   Coexistence of firing rate code and synchronous code in neural populations with synaptic learning.
   International Symposium on Complexity Modelling and its Applications.
   University of Tokyo, Tokyo, Japan, December 6–8 (2004).

5. Naoki Masuda, Kazuyuki Aihara. Band-pass filtering properties of interspike interval encoding with Morris-Lecar neurons.
   Brazilian Symposium on Artificial Neural Networks (SBRN2004).
   Sao Luis, Brazil, September 28 – October 1 (2004).

4. Naoki Masuda, Norio Konno, Kazuyuki Aihara.
   A model of SARS transmission in dynamical small-world networks.
   International Symposium on Dynamical Systems Theory and its Applications to Biology and Environmental Sciences.
   Hamamatsu, Japan, March 14–17 (2004).
   In: Proceedings p.91.

3. Naoki Masuda.
   When can trial averaging of spike trains substitute their population averaging?
   NBNI2003.
   KAIST, Daegeon, South Korea, November 21–22 (2003).

2. Kazuyuki Aihara, Naoki Masuda.
   Spatio-temporal dynamics and duality of rate coding and temporal coding in a neural network model.
   BIOCOMP2002.
   Vietri Sul Mare, Italy, June 3–9 (2002).
   In: Proceedings p.13.

1. Naoki Masuda, Kazuyuki Aihara.
   Spatio-temporal spike encoding of external signals by downstream Morris-Lecar neurons.
   BIOCOMP2002.
   Vietri Sul Mare, Italy, June 3–9 (2002).
   In: Proceedings pp. 119–120.

Poster Presentations (without refereed proceeding papers)

92. Bisna Mary Eldo, Naoki Masuda.
    Are node-activity models on networks graphons?
    NetSci 2026, Boston, MA, USA, 6/1/2026–6/5/2026.

91. Shilong Yu, Weiqi Chu, Naoki Masuda.
    Bounded-confidence models with two-state node activations.
    NetSci 2026, Boston, MA, USA, 6/1/2026–6/5/2026.

90. Fathima Nuzla Ismail, Saiful Islam, Alber Aqil, Omer Gokcumen, Naoki Masuda.
    Multi-layer co-expression networks: Insights into gene expression variability and regulatory interactions.
    Eighth Northeast Regional Conference on Complex Systems.
    Binghamton University, Vestal, NY, 4/9/2025–4/11/2025

89. Harrison Hartle, Naoki Masuda.
    Autocorrelation properties of temporal networks governed by dynamic node-variables.

NetSci 2024,
Quebec City, Canada, 6/16/2024–6/21/2024

88. Saiful Islam, Alber Aqil, Ibrahim Numanagić, Omer Gokcumen, Naoki Masuda.
Exploring genomic evolution through network analysis of duplications across multiple species.
NetSci 2024,
Quebec City, Canada, 6/16/2024–6/21/2024

87. Alber Aqil, Yanyan Li, Zhiliang Wang, Saiful Islam, Madison Russel, Theodora Kunovac Kallak, Marie Saitou, Omer Gokcumen, Naoki Masuda.
Switch-like gene expression modulates disease risk.
Biology of Genomes,
Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, 5/7/2024–5/11/2024

86. Saiful Islam, Alber Aqil, Ibrahim Numanagić, Omer Gokcumen, Naoki Masuda.
Toward understanding genomic segmental duplications by network analysis across multiple species.
NERCCS 2024: Seventh Northeast Regional Conference on Complex Systems,
Clarkson University, Potsdam, NY, 3/20/2024–3/22/2024

85. Neil G. MacLaren, Baruch Barzel, Naoki Masuda.
Approximating nonlinear dynamics on networks by sentinel nodes.
International Symposium on Nonlinear Science and Medicine (ISNSM2024).
University of Tokyo, Tokyo, Japan, 3/15/2024

84. Si Thu Aung, Takamitsu Watanabe, Pitambar Khanra, Naoki Masuda.
Dynamic functional connectivity in patients with major depression.
UB (University at Buffalo) Health AI Symposium.
State University of New York at Buffalo, Buffalo, NY, USA, 2/27/2024

83. Jiyoung Kang, Naoki Masuda.
Quantifying concurrency of edges in temporal networks.
2023 Korean Physics Society Fall Meeting.
Changwon Exhibition Convention Center, Changwon, South Korea, 10/24/2023–10/27/2023

82. Jnanajyoti Bhaumik, Naoki Masuda.
Fixation dynamics for switching networks.
Society for Mathematical Biology Annual Meeting 2023.
Ohio State University, Columbus, OH, 7/16/2023–7/21/2023

81. Kazuki Nakajima, Ruodan Liu, Kazuyuki Shudo, Naoki Masuda.
Quantitative analysis of gender imbalance in East Asian academia.
ICSSI 2023: International Conference on Science of Science and Innovation.
Northwestern University, Evanston, IL, 6/26/2023–6/28/2023

80. Jnanajyoti Bhaumik, Naoki Masuda.
Fixation probability of switching temporal networks.
NERCCS 2023: Sixth Northeast Regional Conference on Complex Systems.
Clarkson University, Potsdam, NY, 3/22/2023–3/24/2023.

79. Madison Russell, Marie Saitou, Omer Gokcumen, Naoki Masuda.
Co-expression networks across different organs reveal physically clustered gene communities.
NERCCS 2023: Sixth Northeast Regional Conference on Complex Systems.
Clarkson University, Potsdam, NY, 3/22/2023–3/24/2023.

78. Pitambar Khanra, Naoki Masuda, Takamitsu Watanabe.
Reliability of energy landscape analysis of resting-state functional MRI data.
Conference on Complex Systems.
Palma de Mallorca, Spain, October 17–21, 2022.

77. Madison Russell, Marie Saitou, Naoki Masuda.
Communities in gene co-expression networks across different organs.
NetSci 2022.
Online, July 25–29, 2022.

76. Maisha Islam Sejunti, Naoki Masuda, Dane Taylor.
Floquet theory for spreading dynamics over periodically switching networks.
NetSci 2022.
Online, July 25–29, 2022.

75. Prosenjit Kundu, Hiroshi Kori, Naoki Masuda.
Validity of a one-dimensional reduction of dynamical systems on networks.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, March 30 – April 1, 2022.
[Honorable Mention of the Best Poster Award]

74. Yanyan Li, Ziliang Wang, Marie Saitou, Naoki Masuda.
Finding genes with outstanding co-expression patterns among organs.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, March 30 – April 1, 2022.

73. Lingqi Meng, Naoki Masuda.
Perturbation theory for graph surgery: A case of theory of cooperation.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, March 30 – April 1, 2022.

72. Madison Russell, Marie Saitou, Naoki Masuda.
Communities in gene co-expression networks across different organs.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, March 30 – April 1, 2022.

71. Chanon Thongprayoon, Lorenzo Livi, Naoki Masuda.
Trajectories of temporal social contact networks.
NERCCS 2022: Fifth Northeast Regional Conference on Complex Systems.
Buffalo, March 30 – April 1, 2022.

70. Noha Abdel-Mottaleb, Kashin Sugishita, Naoki Masuda, Qiong Zhang.
Bio-inspired water distribution network design.
NERCCS 2021: Fourth Northeast Regional Conference on Complex Systems.
Online, March 31 – April 2, 2021.

69. Rupam Acharyya, Penghang Liu, A. Erdem Sariyüce, Naoki Masuda.
Detecting fraudulent users in online marketplaces using temporal motifs.
NERCCS 2021: Fourth Northeast Regional Conference on Complex Systems.
Online, March 31 – April 2, 2021.

68. Ruodan Liu, Masaki Ogura, Elohim Fonseca Dos Reis, Naoki Masuda.
Modeling epidemic spreading in Markovian temporal networks with different degrees of concurrency.

NERCCS 2021: Fourth Northeast Regional Conference on Complex Systems.
Online, March 31 – April 2, 2021.

67. Penghang Liu, Tomomi Kito, Naoki Masuda, A. Erdem Sarıyüce.
Temporal motifs in patent opposition and collaboration networks.
NERCCS 2021: Fourth Northeast Regional Conference on Complex Systems.
Online, March 31 – April 2, 2021. [Best Poster Award]

66. Kazuki Nakajima, Kazuyuki Shudo, Naoki Masuda.
] Configuration models for hypergraphs preserving local quantities of nodes and hyperedges.
NERCCS 2021: Fourth Northeast Regional Conference on Complex Systems.
Online, March 31 – April 2, 2021.

65. Kashin Sugishita, Noha Abdel-Mottaleb, Qiong Zhang, Naoki Masuda.
A growth model for water distribution networks with loops.
COMPLEX20 (Conference on Complex Systems 2020).
Online, December 4–11, 2020.

64. Xiuxiu Zhan, Ziyu Li, Naoki Masuda, Petter Holme and Huijuan Wang.
Spreading-based network embedding for link prediction in static and temporal networks.
NetSci 2020.
Rome, Italy (held online due to COVID-19), September 17–25, 2020.

63. Kashin Sugishita, Mason A. Porter, Mariano Beguerisse-Díaz, Naoki Masuda.
Opinion dynamics in tie-decay temporal networks.
Third Northeast Regional Conference on Complex Systems.
Online, April 1–3, 2020.

62. Lingqi Meng, Naoki Masuda.
Stationary distribution and spectral gap of the node2vec random walks.
Third Northeast Regional Conference on Complex Systems.
Online, April 1–3, 2020.

61. Elohim Fonseca dos Reis, Aming Li, Naoki Masuda.
Human dynamics as a mixture of Poisson processes.
NetSci-X 2020.
Tokyo, Japan, January 20–23, 2020.

60. Lingqi Meng, Naoki Masuda.
Second-order random walks in a toy network.
International Conference on AI and Data Science: Mathematics and Applications.
Suwon, Korea, November 4–5, 2019.

59. Dominik Krzemiński, Naoki Masuda, Khalid Hamandi, Krish Singh, Jiaxiang Zhang.
MEG energy landscape abnormalities in juvenile myoclonic epilepsy.
2019 Conference on Cognitive Computational Neuroscience.
Berlin, Germany, 9/13/2019–9/16/2019.

58. Elohim Fonseca dos Reis, Mark Viney, Naoki Masuda.
Network structure of wild and laboratory mouse immune systems. Threshold Networks. University of Nottingham, Nottingham, UK, July 22–24, 2019.

57. Sadamori Kojaku, Naoki Masuda.
Constructing networks from correlation matrices: An application to economical data.
Threshold Networks. University of Nottingham, Nottingham, UK, July 22–24, 2019.

56. Mengqiao Xu, Qian Pan, Haoxiang Xia, Naoki Masuda.
Estimating international trade statuses of individual countries from a global liner shipping network.
NetSci 2019, University of Vermont, Burlington, VA, USA, May 27–31, 2019.

55. Shun Kodate, Ryusuke Chiba, Shunya Kimura, Naoki Masuda.
Detecting problematic transactions in a consumer-to-consumer e-commerce network.
NetSci 2019, University of Vermont, Burlington, VA, USA, May 27–31, 2019.

54. Sadamori Kojaku, Naoki Masuda.
A generalised significance test for individual communities in networks.
NetSci 2018.
Paris, France, June 11–15 (2018).

53. Koji Oishi, Takamitsu Watanabe, Naoki Masuda.
Overlapping communities in correlation matrices.
NetSci 2018.
Paris, June 11–15 (2018).

52. Christelle van Antwerpen, Giovanni DeMarco, Angela Davies-Smith, Rosemary Jones, Naoki Masuda, N. Jade Thai.
Neural correlates of fatigue: a voxel-based morphometric MRI study in multiple sclerosis.
Organization of Human Brain Mapping Annual Meeting.
Singapore, June 17–21 (2018).

51. Christelle van Antwerpen, N. Jade Thai, Simon Godwin, Angela Davies-Smith, Rosemary Jones, Naoki Masuda.
Impact of fatigue on functional connectivity in multiple sclerosis.
Organization of Human Brain Mapping Annual Meeting.
Singapore, June 17–21 (2018).

50. Simon Godwin, Jade Thai, Christelle van Antwerpen, Lucia Marucci, Angela Davies-Smith, Rosemary Jones, Naoki Masuda.
Resting-state large scale functional networks of patients with multiple sclerosis.
Organization of Human Brain Mapping Annual Meeting.
Singapore, June 17–21 (2018).

49. Sadamori Kojaku, Naoki Masuda.
Multi-scale organisation of core-periphery structure in networks.
1st Laten American Conference on Complex Networks.
Puebla, Mexico, September 25–29 (2017).

48. Masanori Takezawa, Yutaka Horita, Takahiro Ezaki, Naoki Masuda.
What governs behavior in a public goods game: social preferences or reinforcement learning?
CES 2017 (Inaugural Meeting of the Cultural Evolution Society).
Jena, Germany, 9/13/2017–9/15/2017.

47. Sadamori Kojaku, Naoki Masuda.
Core-periphery structure of networks: Consideration for random heterogeneous networks.
NetSci 2017.
Indianapolis, Indiana, USA, June 19–23 (2017).

46. Takahiro Ezaki, Michiko Sakaki, Takamitsu Watanabe, Naoki Masuda.
Energy landscape analysis of age-related changes in the human brain.
Crossroads in Complex Systems. IFISC, Palma de Mallorca, Spain June 5–8 (2017).

45. Naoki Masuda, Takamitsu Watanabe.
Speed of brain dynamics: comparison between autistic and healthy individuals using fMRI data.
Computational Neurology 2017.
Newcastle University, Newcastle, UK, February 20–21 (2017).

44. Sadamori Kojaku, Naoki Masuda.
An extension of modularity for finding multiple core/periphery structure in networks.
NetSci-X 2017.
Tel Aviv, Israel, January 15–18 (2017).
[2017 NetSci-X Best Contribution on Financial Networks Award, sponsored by The Journal of Network Theory in Finance]

43. Takahiro Ezaki, Yutaka Horita, Masanori Takezawa, Naoki Masuda.
Reinforcement learning accounts for conditional cooperation behavior in social dilemma games.
Conference on Complex Systems.
Amsterdam, Netherland, September 19–22 (2016).

42. Takahiro Ezaki, Yutaka Horita, Masanori Takezawa, Naoki Masuda.
Reinforcement learning explains conditional cooperation in repeated social dilemma games. Asia-Pacific Econophysics Conference 2016.
University of Tokyo, Tokyo, Japan, 8/24/2016–8/26/2016.
[Best Poster Award]

41. Yutaka Horita, Masanori Takezawa, Takuji Kinjo, Yo Nakawake, Naoki Masuda.
Transient nature of pay-it-forward reciprocity.
The 31st International Congress of Psychology.
PACIFICO Yokohama, Yokohama, Japan, 7/24/2016–7/29/2016.

40. Yutaka Horita, Masanori Takezawa, Takuji Kinjo, Yo Nakawake, Naoki Masuda.
Contagion of cooperation in a donation game played on chain networks.
NetSci 2016.
Seoul, South Korea, 5/30/2016–6/3/2016.

39. Masato S. Abe, Naoki Masuda.
Modeling dominance networks with intrinsic node weights.
Conference on Complex Systems.
Tempe, Arizona, USA, 9/28/2015–10/2/2015.

38. Masato S. Abe, Naoki Masuda.
Analysis of dominance network models with intrinsic node weights.
2015 Joint Meeting of JSMB and CJK Colloquium on Mathematical Biology, Doshisha University, Kyoto, Japan, August 26–29 (2015).

37. Leo Speidel, Taro Takaguchi, Kazuyuki Aihara, Naoki Masuda.
Community detection in directed acyclic graphs.
NetSci 2015.
Zaragoza, Spain, June 1–5 (2015).

36. Takamitsu Watanabe, Naoki Masuda, Fukuda Megumi, Ryota Kanai, Geraint Rees. Energy landscape of human brain activity during bistable perception.
Society for Neuroscience.
Washington DC, November 15–19 (2014).

35. Leo Speidel, Renaud Lambiotte, Kazuyuki Aihara, Naoki Masuda.
Active and passive random walks on stochastic temporal networks.
Social Modeling and Simulations + Econophysics Colloquium 2014.
Nichii Gakkan, Kobe, Japan, November 4–6 (2014).

34. Takehisa Hasegawa, Taro Takaguchi, Naoki Masuda.
Network observability transitions under degree correlation.
NetSci 2014.
Berkeley, CA, USA, June 2–6 (2014).

33. Ryosuke Nishi, Naoki Masuda.
Dynamics of social balance on the temporal complete graph.
NetSci 2014.
Berkeley, CA, USA, June 2–6 (2014).

32. Leo Speidel, Renaud Lambiotte, Kazuyuki Aihara, Naoki Masuda.
Passive random walks on stochastic temporal networks.
NetSci 2014.
Berkeley, CA, USA, June 2–6 (2014).

31. Takehisa Hasegawa, Taro Takaguchi, Naoki Masuda.
Impact of correlated structure on network observability transition.
The 3rd International Symposium on Innovative Mathematical Modelling.
University of Tokyo, Tokyo, Japan, November 12–15 (2013).

30. Ryosuke Nishi, Naoki Masuda.
Imitation of immunization behavior on scale-free networks under different update rules.
International Workshop on Phase Transition, Critical Phenomena and Related Topics in Complex Networks.
Hokkaido University, Sapporo, Japan, September 9–11 (2013).

29. Taro Takaguchi, Nobuo Sato, Kazuo Yano, Naoki Masuda.
Inferring directed static networks of influence from undirected temporal networks.
The 37th Annual International Computer Software & Applications Conference (COMPSAC).
Kyoto, Japan, July 22–26 (2013).

28. Ryosuke Nishi, Naoki Masuda.
Bayesian collective opinion formation under confirmation bias.
25th Conference on Statistical Physics of the International Union for Pure and Applied Physics (STATPHYS 25).
Seoul National University, Seoul, South Korea, July 22–26 (2013).

27. Takamitsu Watanabe, S. Hirose, H. Wada, Y. Imai, T. Machida, I. Shirouzu, S. Konishi, Y. Miyashita, Naoki Masuda.
Second-order maximum entropy models applied to macroscopic resting-sate networks in human brains.
NetSci 2013.
Copenhagen, Denmark, June 3–7 (2013).

26. Taro Takaguchi, Nobuo Sato, Kazuo Yano, Naoki Masuda.
Extracting directed influential relationship between nodes from undirected temporal networks.
NetSci 2013.
Copenhagen, Denmark, June 3–7 (2013).

25. Mitsuhiro Nakamura, Naoki Masuda.
Stern moral assessment promotes ingroup favoritism in indirect reciprocity.
8th Conference on European Social Simulation Association.
Salzburg, Austria, September 10–14 (2012).

24. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Structure of cell networks critically determines oscillation regularity.
International Symposium on Complex Systems 2011.
University of Tokyo, Tokyo, Japan, December 1–3 (2011).

23. Ryo Fujie, Naoki Masuda, Kazuyuki Aihara.
Modeling competition of more than two languages.
International Symposium on Complex Systems 2011
University of Tokyo, Tokyo, December 1–3 (2011).

22. Taro Takaguchi, Mitsuhiro Nakamura, Nobuo Sato, Kazuo Yano, Naoki Masuda.
Predictability of conversation partners in Japanese company offices.
International Symposium on Complex Systems 2011
University of Tokyo, Tokyo, Japan, December 1–3 (2011).

21. Taro Takaguchi, Mitsuhiro Nakamura, Nobuo Sato, Kazuo Yano, Naoki Masuda.
Predictability of communication patterns and its correlates to individuals' positions in social networks.
European Conference on Complex Systems (ECCS'11).
Vienna, Austria, September 12–16 (2011).

20. Taro Takaguchi, Naoki Masuda.
Effect of non-Poissonian inter-event intervals on opinion dynamics.
NetSci 2011.
Budapest, Hungary, 6/8/2011–6/10/2011.

19. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Collective enhancement of regularity in networks of noisy networks.
Far-From-Equilibrium Dynamics 2011.
Kyoto University, Kyoto, Japan, January 4–8 (2011).

18. Ralf Toenjes, Naoki Masuda, Hiroshi Kori.
Synchronization transition of identical phase oscillators in a directed small-world network.
Far-From-Equilibrium Dynamics 2011.
Kyoto University, Kyoto, Japan, January 4–8 (2011).

17. Naoki Masuda.
A method to immunize networks with community structure.
NetSci 2010.
Boston, MA, USA, May 10–14 (2010).

16. Yoshimi Yoshino, Naoki Masuda.
Evolutionary Prisoner's Dilemma on dynamic networks: results for different payoff schemes.
BIFI2010 International Congress (Networks: A framework for cross disciplinary applications).
Zaragoza, Spain, February 3–6 (2010).

15. Hiroshi Kori, Yoji Kawamura, Naoki Masuda.
Relation between fluctuations in isolated components and those in networked components.

International Symposium on Complex Systems Biology.
University of Tokyo, Tokyo, Japan, Septermber 29 – October 1 (2009).

14. Jin Seop Kim, Naoki Masuda, Byungnam Kahng.
A priority queue model with bursty arrivals of incoming tasks.
International Conference on Complex Networks: The Past 10 Years and Future.
Seoul, South Korea, 12/19/2008–12/22/2008.

13. Naoki Masuda, Brent Doiron.
Signal discrimination performed by population of spiking neurons enhanced by background gamma oscillations.
The Annual Computational Neuroscience Meeting (CNS2008).
Portland, Oregon, USA, July 19–24 (2008).

12. Taro Ueno, Naoki Masuda.
Effective intervention and vaccination strategies against nosocomial infection based on network analysis.
13th International Society for Infectious Diseases.
Kuala Lumpur, Malaysia, June 19-22 (2008).
In: International Journal of Infectious Diseases. 12 (Supplement 1), E347–E348 (2008).

11. Naoki Masuda, Shun-ichi Amari.
Modeling memory transfer in cerebellar vestibulo-ocular-reflex learning.
The Annual Computational Neuroscience Meeting (CNS2006).
Edinburgh, UK, 7/16/2006–7/20/2006.

10. Michael Gutmann, Naoki Masuda, Fusao Kato, Kazuyuki Aihara.
Bursting pacemaker models for the preBotzinger complex.
Japan-Germany Symposium on Computational Neuroscience.
RIKEN Brain Science Institute, Wako, Japan, 2/1/2006–2/4/2006.

9. Michael Gutmann, Naoki Masuda, Kazuyuki Aihara.
Bifurcations in respiratory neuronal pacemaker models.
International Symposium on Complexity Modelling and its Applications 2005.
University of Tokyo, Tokyo, Japan, November 21–23 (2005).

8. Kazumichi Ohtsuka, Norio Konno, Naoki Masuda, Kazuyuki Aihara.
Phase transition and indirect effects of the four-state stochastic model on the square lattice.
European Conference on Mathematical and Theoretical Biology (ECMTB) 2005.
Dresden, Germany, 7/18/2005–7/22/2005.
In: Proceedings, v.2, p.277.

7. Naoki Masuda, Masato Okada, Kazuyuki Aihara.
Spatial filtering by locally connected neurons.
The Annual Computational Neuroscience Meeting (CNS2005).
Madison, Wisconsin USA, July 17–21 (2005).

6. Kazumichi Ohtsuka, Norio Konno, Naoki Masuda, Kazuyuki Aihara.
Phase transition and coexistence of three-state stochastic models on the two-dimensional lattice space.
5th International Conference on Biological Physics (ICBP 2004).
Gothenburg, Sweden, August 23–27 (2004).

5. Naoki Masuda, Kazuyuki Aihara.
   Encoding multiple temporal waveforms by neural population with spike-time-dependent plasticity.
   The Annual Computational Neuroscience Meeting (CNS2004).
   Baltimore, Maryland, USA, July 17–22 (2004).

4. Naoki Masuda, Hiroyoshi Miwa, Norio Konno.
   Emergence of power law in dynamical thresholding network models with intrinsic vertex weights.
   International Conference on Statistical Physics (Statphys Kolkata V).
   Kolkata, India, 6/27/2004–7/1/2004.

3. Naoki Masuda, Kazuyuki Aihara.
   Dual coding and effects of global feedback in multilayered neural networks.
   The Annual Computational Neuroscience Meeting (CNS2003).
   Alicante, Spain, July 5–9 (2003).

2. Naoki Masuda, Kazuyuki Aihara.
   Dual coding and clustering in feedforward neural networks.
   The 2003 Latsis Symposium on Neural Coding and Modeling.
   EPFL, Lausanne, Switzerland, February 17–19 (2003).

1. Naoki Masuda, Kazuyuki Aihara.
   Population rate coding and synchrony-based temporal coding dually appearing in feedforward neural networks.
   7th Tamagawa Dynamic Brain Forum.
   Visegrad, Hungary, September 8-11 (2002).
   In: Proceedings p.36.

Outreach Activity Presentations

3. Dynamic programming.
   The Michigan Math Circle.
   University of Michigan, MI, 2/19/2026
   [Audience was middle and high -school students.]

2. Network science: Mathematical modeling and interdisciplinary applications.
   WNY (Western New York) STEM Forum.
   State University of New York at Buffalo, NY, 10/9/2024
   [A main body of audience was local teachers.]

1. Mathematical biology presentation and panel discussion.
   WiSE (Women in Science and Engineering) STEM Outreach Campus Day.
   State University of New York at Buffalo, NY, 4/26/2024
   [Audience was local female highschool students.]

Professional Service

Editorial Work

Regular

9/2025–present	Editorial Board, Journal of the Royal Society Interface
12/2024–present	Associate Editor, SIAM Journal on Applied Mathematics
11/2023–present	Editor, Journal of Computational Science
10/2022–present	Associate Editor, Frontiers in Complex Systems, Complex Networks section
6/2020–present	Associate Editor, Frontiers in Physics, Social Physics section
6/2019–present	Editor, Journal of Computational Social Science
1/2019–present	Associate Editor, Journal of Complex Networks
10/2020–5/2026	Academic Editor, Complexity
1/2014–1/2025	Editorial Board, Scientific Reports
3/2018–2/2023	Editorial Board, Journal of Physics Communications
5/2014–11/2023	Editorial Board, PLoS ONE
6/2013–12/2019	Review Editor, Frontiers in Physics, Interdisciplinary Physics section

Ad-hoc

2026	Guest Editor (for one submission), PLoS Digital Health
2020	Guest Editor, Japanese Economic Review
	Special Issue: Economics and Complex Networks (co-edited with Teruyoshi Kobayashi)

Organization of Conferences and Workshops

Organizer (co-organized with Weiqi Chu and Ying-Cheng Lai)
Dimension reduction for network-related dynamics.
The Institute for Mathematical and Statistical Innovation (IMSI) workshop.
Chicago, IL, 6/21/2027–6/25/2027.

Proceedings Program Committee Member
ISMB (Intelligent Systems for Molecular Biology) 2026

Organizer (co-organizerd with Quanjun Lang)
Minisymposium: Data-driven Learning and Reduced Order Modeling of Interacting Particle Systems.
SIAM Conference on Uncertainty Quantification (UQ26),
Minneapolis, MI, 3/22/2026

Program Committee Member
International Conference on Complex Systems Modeling, Analysis & Applications 2025, 2026

Program Committee Member
NetSci 2013, 2016 (Program Co-Chair), 2017, 2018 (Senior Program Committee Member), 2019, 2020, 2022, 2023, 2024, 2025, 2026

Program Committee Member
NetSci-X 2015, 2016, 2017, 2018, 2020 (Program Chair), 2022, 2023, 2024, 2026

Program Committee Member
Northeast Regional Conference on Complex Systems, 2020 (also Poster Chair), 2021 (General Co-Chair), 2022 (General Co-Chair), 2025

Program Committee Member
CompleNet 2019, 2020, 2021, 2023, 2024, 2025

International Advisory Committee Member
28th IUPAP International Conference on Statistical Physics (STATPHYS28)
Tokyo, Japan, 8/7/2023–8/11/2023

Vice Chair
2022 IEEE International Conference on Big Data

Osaka, Japan, 12/17/2022–12/20/2022

Program Committee Member
Conference on Complex Systems 2022, 2026 (Program Chair)

Organizer (Coorganized with Tomomi Kito and Kazuki Nakajima)
Socioeconomic Networks and Network Science Workshop 2022
Online, 8/1/2022–8/2/2022

Organizer (Coorganized with Tomomi Kito and Kazuki Nakajima)
Science of Innovation and Success Workshop
Waseda University, Japan (hybrid format), 8/2/2021–8/3/2021

Poster Chair
Tenth International Conference on Complex Systems (ICCS 2020)
Online, 7/27/2020–7/31/2020

Organizer (Coorganized with Tomomi Kito and Yuka Fujiki)
Socioeconomic Networks and Network Science Workshop
Waseda University, Japan (held online due to COVID-19), 7/3/2020–7/4/2020

Program Committee Member
8th Annual SIAM Workshop on Network Science (NS20)
Toronto, Ontario, Canada, 7/9/2020–7/10/2020

Organizer (Coorganized with Guido Caldarelli, Hiroyasu Inoue, and Carolina Mattsson)
NetSci-X Special Workshop on Economic & Financial Networks.
Tokyo, Japan, 1/23/2020

Program Committee Member
International Conference on Social Informatics 2017, 2018, 2019, 2020 (Senior Program Committee Member)

Program Committee Member
Workshop on Social Influence 2014, 2016, 2017, 2018

Program Committee Member
Annual International Conference on Computational Social Science 2018, 2019, 2020

Organizer (Coorganized with Nicola Perra and Vincenzo Nicosia)
2nd UK Network Science Workshop
University of Greenwich, London, UK, 10/26/2018

Program Committee Member
SIAM Workshop on Network Science 2018 (SIAMNS 2018)
Portland, OR, USA, 7/12/2018–7/13/2018

Program Committee Member
Integration of Empirical Data in Network Epidemiology (iodine)
A satellite symposium in NetSci 2018, Paris, France, 6/12/2018

Organizer
UK Network Science Workshop
University of Bristol, Bristol, UK, 5/11/2018

Program Committee Member
International Workshop on Complex Networks and Their Applications, 2014, 2015, 2016, 2017, 2018, 2019

Organizer
Engineering Applications of Networks Workshop
University of Bristol, Bristol, UK, 11/1/2017

Program Committee Member
CCS2017 Satellite Meeting Modeling of Disease Contagion Processes 6th Edition
Cancun, Mexico, 9/17/2017–9/22/2017

Program Committee Member
4th European Network Intelligence Conference 2017 (ENIC'17)
Duisburg, Germany, 9/11/2017–9/12/2017.

Organizer (coorganized with Juyong Park)
Satellite workshop: Competition networks and centrality
NetSci 2016
Seoul, Korea, 5/30/2016

Organizer (coorganized with Byungnam Kahng and Zengru Di)
Satellite workshop: Network science research in Asia: Fundamentals and applications
NetSci 2016
Seoul, Korea, 5/30/2016–5/31/2016

Coorganizer
Special Session: Epidemic dynamics: mathematical modeling and data analysis
4th IFAC Conference on Analysis and Control of Chaotic Systems
Tokyo, Japan, 8/26/2015–8/28/2015

Program Committee Member
Second International Workshop on Complex Networks and their Applications
Kyoto, Japan, 12/2/2013–12/5/2013

Program Committee Member
First Workshop on Rational, Secure and Private Ad-hoc Networks (RASEP'11)
Crete, Greece, 10/16/2011–10/21/2011

Program Committee Member
The 9th Asia-Pacific Complex Systems Conference (Complex'09)
Tokyo, Japan, 11/4/2009–11/7/2009

Program Committee Member
Special Session: Evolutionary Games on Complex Networks
IEEE Congress on Evolutionary Computation (CEC 2009)
Trondheim, Norway, 5/18/2009–5/21/2009

Coorganizer
NSC Winter Workshop 2008: Complex Nonlinear Dynamics ranging from Biology to Engineering
Sapporo, Japan, 3/8/2008–3/10/2008

Organizer (Coorganized with Martin Hasler, Kazuyuki Aihara, and Oscar De Feo)
Special Session: Nonlinear Analysis and Modeling of Neural Signal Processing and Coding
International Symposium on Nonlinear Theory and its Applications (NOLTA2004)
Fukuoka, Japan, 11/29/2004–12/3/2004

Award Judge

Japanese Society for Mathematical Biology Early Career Award, 2019 (Head of the Committee), 2022, 2023.

10/2020–09/2022 Judge for Akira Okubo Prize,
jointly administered by the Society for Mathematical Biology and the Japanese Society for Mathematical Biology.

Member of Erdős-Rényi Prize Committee, 2020, Network Science Society.

Young Scientist Award in Socio- and Econophysics 2020 and 2025
Division of socio-economic physics, German Physical Society.

Referee for Grant Applications

National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) Special Emphasis Panel, ZDK1 GRB-S (M2), member, 2/2025

Army Research Office

Air Force Office of Scientific Research

National Science Foundation (panelist)

Concerted Research Actions (CRA) 2024, A university in Belgium

Research Foundation Flanders (FWO), Belgium, including FWO Review College panel membership in 2023–2028

Austrian Science Fund (FWF)

Swiss National Science Foundation

Engineering and Physical Sciences Research Council (EPSRC), UK

National Science Centre, Poland

Fund for Scientific Research-FNRS, Belgium ...

Programma per Giovani Ricercatori – Rita Levi Montalcini, Italy

Japan Society for the Promotion of Science, Japan

MATH AmSud, cooperation between France and Latin America

Referee for Peer-Reviewed Journals

More than 100 journals including Nature Communications, Nature Physics, Physical Review Letters, Physical Review X, Proceedings of the National Academy of Sciences of the United States of America, Science Advances, and SIAM Review.
For a full list, see naokimas.github.io/naokimasuda.github.io/publ_e.html.

Membership in Invitation-Based Organizations

10/2012–present H1Connect (formerly, Faculty Opinions and F1000 Faculty) Member

Membership in Other Professional Organizations

Network Science Society

Society for Industrial and Applied Mathematics (SIAM)

American Mathematical Society

Society for Mathematical Biology

American Physical Society

Japanese Society for Mathematical Biology

Enterprise Partnership

3/2023–12/2023	Subject Matter Expert, Freelancer Technology Pty Limited (Project with Logistic Management Institute (LMI), Center for Disease Control and Prevention (CDC), and National Aeronautics and Space Administration (NASA))
10/2021–6/2022	Subject Matter Expert, Freelancer Technology Pty Limited (Project with Center for Disease Control and Prevention (CDC) and National Aeronautics and Space Administration (NASA))
4/2020–present	Lecture speaker (in Japanese) on academic writing, Enago
12/2018–present	Technical Advisor, Nodes & Links Limited

Evaluator for Promotion and Tenure for Other Institutions

Eight times for R1 universities and one time for an R2 university in the US

One Russell Group University in the UK

One double first-class construction university in China

One university in Japan

One university in South Korea

University Service

University of Michigan

2/2026–4/2026: REU (Research Experiences for Undergraduates) Committee member, Department of Mathematics

11/2025–2/2026: committee member, Department of Computational Medicine and Bioinformatics PhD Admissions Committee

State University of New York at Buffalo

12/2024–3/2025: committee member, 2024-25 Excellence in Teaching Award for Graduate Teaching Assistants, Graduate School

12/2023–6/2025: SUNY Office of Research, Innovation and Economic Development Reviewer Panel (This is a State University of New York systemwide service.)

9/2023–3/2024: member, Graduate School Executive Committee sub-committee on International Graduate Student Language Assessment

7/2023–1/2024: Chair of the Applied Mathematics Hiring Committee, Department of Mathematics

2023: Ad-hoc committee for faculty's promotion, Department of Mathematics

4/2023–6/2025: Diversity, Equity and Inclusive Committee, Department of Mathematics

2023–6/2025: faculty advisor, American Mathematical Society graduate student chapter at University at Buffalo

2022–2023: Hiring Committee for the External Chair, Department of Mathematics

7/2022–6/2025: Graduate School Director, Department of Mathematics

2022–2023: Executive Committee, Department of Mathematics

1/2022–2/2022: Ad-hoc Committee for graduate student admission for Computational and Data Enabled Science and Engineering (CDSE) Program

2021–2022: Applied Mathematics Hiring Committee, Department of Mathematics

2021: Ad-hoc Committee for the Creation of a Departmental Disciplinary Excellence Proposal

2020, 2021: Graduate Studies Awards Committee

2020: Ad-hoc Committee for Faculty's Reappointment, Department of Mathematics

2020: Chairmanship Committee, Department of Mathematics

2019: Applied Mathematics Hiring Committee, Department of Mathematics

2019–2025: Department of Mathematics Area Committee in Applied Mathematics

2019–2025: Department of Mathematics Graduate Studies Committee

University of Bristol, United Kingdom

10/2018–8/2019: Year 2 (Undergraduate) Tutor

10/2016–8/2019: Study Abroad Academic Director

9/2014–8/2019: Personal Tutor (≈ 20 undergraduate and master course students each year)

University of Tokyo, Japan (all roles are officially in Japanese; translation by NM)

2012: Curriculum Coordinator (Graduate)

2012: Coordinator for Departmental Brochures

2011: Committee Member for the Tenth Anniversary Event of the Graduate School

2008-2011: Curriculum Coordinator (Undergraduate)

2008-2011: Library Committee Member

2007, 2008, 2010, 2012: Organizer, Colloquium on Mathematical Informatics

2007: Minute taker

2007: Coordinator of Factory Tours (Undergraduate)

Teaching

University of Michigan

MATH 463: Mathematical Biology (for undergraduate students) 3 credit hours
Fall 2026

MATH 462: Mathematical Modeling (for undergraduate students) 3 credit hours
Winter 2026

State University of New York at Buffalo

MTH 550: Network Theory (for graduate students), also listed as MTH 463 (Spring 2021) and MTH 450 (Fall 2022) for undergraduate students
3 credit hours
Spring 2021, Fall 2022, Spring 2025

MTH 563: Topics in Applied Math: Stochastic Processes on Networks (for graduate students)
3 credit hours

Spring 2020, Spring 2022, Spring 2023

MTH 306: Introduction to Differential Equations (for undergraduate students)
4 credit hours
Spring 2020, Spring 2021, Spring 2022

MTH 537: Introduction to Numerical Analysis I (for graduate students)
3 credit hours
Fall 2019, Fall 2020, Fall 2021

University of Bristol, United Kingdom

EMAT 20920: Numerical Methods in MATLAB (for second year undergraduate)
10 CPs
Autumn 2014, 2015, 2016, and 2017

EMAT 33100: Nonlinear Dynamics and Chaos (for third year undergraduate)
10 CPs
Autumn 2015, Autumn 2016 (50% of teaching)

EMATM001: Advanced Nonlinear Dynamics and Chaos (for masters)
10 CPs
Autumn 2018 (50% of teaching)

EMAT 30670: Optimisation Theory and Applications (for third year undergraduate)
10 CPs
Autumn 2016 (50% of teaching), Autumn 2017 (50%), Autumn 2018 (100%)

EMAT 22220: Mathematical and Data Modelling 2 (for second year undergraduate)
20 CPs
Autumn 2014, 2015, 2016, 2017, 2018
Role: supervision and marking of 3–4 group projects of 4 weeks

EMAT 30005: Mathematical and Data Modelling 3 (for third year undergraduate)
30 CPs
2017/18, 2018/19 (25% of teaching) Role: organization, supervision, and marking of group projects for a semester with other teaching staff

University of Tokyo, Japan

4860-1005: Mathematical Structures in Informatics (for Ph.D. and M.Sc students)
2 credit hours, 1 semester
Spring 2009 and Spring 2011 (in Japanese), Spring 2013 (in English)
Role: Instructor (solo)

03-541530: Topology (3rd year undergrad. Originally in Japanese. Translation by NM)
1.5 credit hours, 1 semester
Fall 2007–2013.
Role: Instructor (solo)

03-541620 Group Journal Club (4th year undergrad. Originally in Japanese. Translation by NM)
1.5 credit hours (co-taught with another faculty), 1 semester
Spring 2007, 2008, 2010, 2011, 2013
Role: Sole unit organizer

Other

Lecturing (on temporal networks)
"Advanced Complex Networks" in the Master of Physics of Complex Systems.

Programa d'Impuls i Internacionalització per a Estudis de Postgrau.
University of Balearic Islands, Spain, 5/29/2017–6/2/2017

Part-time Lecturer
7735: Social Simulations (3rd year undergraduate. Originally in Japanese. Translation by NM)
Tokyo Institute of Technology, Japan
Spring 2008–2013. 2 credit hours

Research Advising

PhD thesis supervision — current

University of Michigan

2/2024–present	Shilong Yu (transferred from SUNY Buffalo, 8/2025) Applied & Interdisciplinary Mathematics (AIM) PhD Program (main supervision by NM, co-advised by Daniel Forger)
8/2023–present	Bisna Mary Eldo (transferred from SUNY Buffalo, 8/2025) Applied & Interdisciplinary Mathematics (AIM) PhD Program (main supervision by NM, co-advised by Victoria Booth)

State University of New York at Buffalo

8/2022–present	Katherine Betz Department of Mathematics (sole supervision by NM)

PhD thesis supervision — completed

State University of New York at Buffalo

4/2021–12/2025	Jnanajyoti Bhaumik Department of Mathematics (sole supervision by NM)
8/2021–8/2025	Saiful Islam Computational and Data-Enabled Sciences PhD Program (sole supervision by NM)
7/2020–2/2025	Chanon Thongprayoon Department of Mathematics (sole supervision by NM)
2/2020–5/2024	Ruodan Liu Department of Mathematics (sole supervision by NM)
8/2019–5/2022	Lingqi Meng Department of Mathematics (sole supervision by NM)
1/2018–5/2022	Elohim Fonseca dos Reis (transferred from University of Bristol, 8/2019) Department of Mathematics (sole supervision by NM)

University of Bristol, United Kingdom

9/2015–8/2019	Simon Godwin Department of Engineering Mathematics (2nd supervisor, 50%; co-supervised with Lucia Marucci)
9/2015–11/2018	Christelle Langley Doctorate in Philosophy (in Cognitive Neuroscience) (2nd supervisor, 20%; co-supervised with Jade Thai)

University of Tokyo, Japan

4/2010–3/2013	Taro Takaguchi Department of Mathematical Informatics (sole supervision by NM)
4/2010–3/2013	Mitsuhiro Nakamura Department of Mathematical Informatics (sole supervision by NM)

PhD student supervision — terminated

State University of New York at Buffalo

8/2023–6/2025 Nurjahan Akter Nepa
Department of Mathematics
(sole supervision by NM;
terminated in the middle because of NM's move to University of Michigan)

8/2023–5/2025 Quoc Chuong Nguyen
Department of Mathematics
(sole supervision by NM;
terminated in the middle because of NM's move to University of Michigan)

8/2021–8/2025 Maisha Islam Sejunti
Department of Mathematics
(co-supervision with Dane Taylor till 8/2023;
co-supervision terminated because DT moved to University of Wyoming;
NM's supervision terminated because of NM's move to University of Michigan;
MIS transferred to University of Wyoming to be advised by DT)

8/2021–12/2023 Madison Russell
Department of Mathematics
(sole supervision by NM;
graduated with a master (MA) degree)

1/2021–6/2021 Luca Sicuso
Department of Mathematics
(sole supervision by NM)

University of Bristol, United Kingdom

4/2019–8/2019 Maximilian Kloucek
Department of Physics
(2nd/3rd supervisor; co-supervised with Paddy Royall)
(terminated in the middle because of NM's move to Buffalo)

9/2017–8/2019 Robert Eyre
Department of Engineering Mathematics)
(20% supervision; co-supervised with Filippo Simini)
(terminated in the middle because of NM's move to Buffalo)

9/2016–8/2019 Alfie Wearn
PhD program in Neural Dynamics
(2nd supervisor, 50%, co-supervised with Elizabeth Coulthard and Risto Kauppinen)
(terminated in the middle because of NM's move to Buffalo)

Master thesis supervision

University of Michigan

7/2026–present Alec Fleur
Applied & Interdisciplinary Mathematics (AIM) MS + PhD Program
(supervision by NM)

9/2025– Anand Parikh
Bioinformatics Master's Program
(sole supervision by NM)

University of Bristol, United Kingdom, Department of Engineering Mathematics

1/2019–8/2019 Daiying Zhu (MSc)
"Project networks: data analysis and mathematical modelling"

1/2018–8/2018 Victor Küpper (MSc)
"Recurrence quantification analysis of recurrence plots with application to MEG data"

1/2018–8/2018 Longxin Wang (MSc)
"Epidemic dynamics modelling of disease progression in Alzheimer disease"

9/2018-5/2019 Arthur Dodson (MEng)
"Network-based early warning signals for financial markets"

9/2017–5/2018 Duncan Cassells (MEng)
"Energy landscape of financial markets"

9/2016–5/2017 Sophie Landon (MEng)
"Analysis and further development of an agent-based model for nest selection in the ant *Temnothorax albipennis*"

9/2016–5/2017 Ryan Stuart (MEng)
"Autistic brain networks"

9/2015–5/2016 Jamie Wright (MEng)
"Network based sports rankings in international football"

9/2014–5/2015 Bobby East (MEng)
"Network-based ranking systems — An analysis of formula 1 driver ability"

University of Tokyo, Japan, Department of Mathematical Informatics, MSc
(The thesis title is in Japanese so not shown.)

4/2012–2/2014 Yuta Ohashi

4/2012–2/2014 Kodai Saito

4/2012–2/2014 Daiki Takeuchi

4/2011–3/2013 Rintaro Ishii

4/2011–3/2013 Hiroaki Tomita

4/2011–3/2013 Shun Motegi

4/2010–3/2012 Takaoki Ohnishi

4/2010–3/2012 Shinya Kato

4/2009–3/2011 Kazuya Ugayama

4/2009–3/2011 Yasushi Kawase

4/2009–3/2011 Ryogo Sekine

4/2008–3/2010 Hiroto Tateishi

4/2007–3/2009 Masashi Tsubosaka

Undergraduate student research supervision

University of Michigan

2/2026– Zhiheng Xian
mathematics major
(joint supervision with Dr. Youngheun Jo, postdoc at Department of Psychiatry)

9/2025–5/2026 Tianyong Yao
mathematics major
(sole supervision by NM)

State University of New York at Buffalo

8/2021–5/2022 Yanyan Li (Senior Thesis)
Department of Mathematics
(sole supervision by NM)

7/2020–5/2021 Mengyuan (Jason) Sun (Senior Thesis)
Department of Mathematics
(sole supervision by NM)

University of Bristol, United Kingdom

9/2016–5/2017 Yiyoung Kim (BEng)
Department of Engineering Mathematics
"Immunising networks"

University of Tokyo, Japan, Department of Mathematical Engineering and Information Physics
(The thesis title is in Japanese so not shown.)

9/2013–2/2014 Yuni Iwamasa

9/2013–2/2014 Jun Kato

9/2012–3/2013 Yusuke Enomoto

9/2012–3/2013 Tomonori Murakoshi

9/2012–3/2013 Chihiro Watanabe

9/2011–3/2012 Kota Ishihara

9/2011–3/2012 Yuki Okano

9/2010–3/2011 Takatoshi Sakakibara

9/2010–3/2011 Shoma Tanabe

9/2010–3/2011 Naotaka Yamada

9/2009–3/2010 Kotaro Itoyama

9/2009–3/2010 Yoshiyuki Oiwa

9/2009–3/2010 Jotaro Doi

9/2008–3/2009 Michiru Akita

9/2008–3/2009 Akio Iwagami

9/2007–3/2008 Yuko Takahashi

9/2007–3/2008 Ken Tsukada

Rotation and summer project (or similar) supervision

State University of New York at Buffalo

1/2024–12/2024 HyunAh Lee
Department of Biostatistics
(effective co-supervision with Ekaterina Noyes, one-year funded project)

University of Bristol, United Kingdom, PhD program in Neural Dynamics

01/2019–04/2019 Arthur Spencer (co-advised with Ela Chakkarapani and Jon Brooks)

05/2018–08/2018 Oscar Davy (co-advised with Jade Thai)

01/2018–04/2018 Anne-Lene Sax (co-advised with Elizabeth Coulthard)

01/2016–04/2016 Alfie Wearn (co-advised with Elizabeth Coulthard and Risto Kauppinen)

05/2015–08/2015 Felicity Inkpen (co-advised with Risto Kauppinen)

Junior faculty mentor

State University of New York at Buffalo

8/2023–8/2025 Sarbendu Rakshit
Department of Mathematics
Visiting Assistant Professor (full time)

1/2021–12/2022 Prosenjit Kundu
Department of Mathematics
Visiting Assistant Professor (full time)

Fellowship sponsor

University of Michigan

7/2026–present Dr. Kai S. Shimagaki
Assistant Research Scientist
(Michigan Center for Applied and Interdisciplinary Mathematics)

University of Bristol, United Kingdom

4/2017–3/2019 Dr. Christos Ellinas
"Application of complex networks to project robustness and resilience"
EPSRC Doctoral Prize Fellowship

Postdoc supervision

University of Michigan

8/2026–	Haridas Kumar Das (funded by Japan Science and Technology Agency)
4/2026–	Yusuke Kato (funded by Japan Science and Technology Agency)
9/2025–3/2026	Alber Aqil (funded by Japan Science and Technology Agency)
8/2024–6/2026	Tharusha Bandara (funded by Japan Science and Technology Agency) (transferred from SUNY Buffalo, 8/2025)

State University of New York at Buffalo

1/2025–11/2025	Fathima Nuzla Ismail (funded by Japan Science and Technology Agency)
5/2023–10/2024	Si Thu Aung (funded by Japan Science and Technology Agency)
1/2022–7/2024	Neil G. MacLaren (funded by Japan Science and Technology Agency)
12/2021–11/2023	Pitambar Khanra (funded by Japan Science and Technology Agency)
8/2020–5/2021	Rupam Acharyya (funded by JP Morgan Chase)
4/2019–3/2021	Kashin Sugishita (funded by Japan Society for the Promotion of Science; transferred from University of Bristol, 08/2019)

University of Bristol, United Kingdom

9/2019–2/2020	Irene Malvestio (funded by Cookpad Limited)
6/2019–1/2020	Marinho Antunes Lopes (funded by GW4, UK)
7/2018–7/2019	Alessio Cardillo (funded by Cookpad Limited)
4/2016–3/2019	Sadamori Kojaku (funded by CREST, Japan Science and Technology Agency (JST))
5/2015–3/2016	Kohei Tamura (funded by CREST, JST)

University of Tokyo, Japan

4/2013–2/2014	Takuya Machida (funded by Research Fellowships for Young Scientists (PD) Japan Society for the Promotion of Science (JSPS))
4/2012–10/2012	Ryosuke Nishi (funded by Kakenhi, Japan)
4/2011–3/2012	Jun Nakabayashi (funded by Kakenhi, Japan)
4/2011–3/2012	Etsuo Segawa (funded by JST, Presto)
4/2011–3/2012	Shuhei Furuya (funded by Research Fellowships for Young Scientists (PD), JSPS)
4/2009–3/2011	Yoshimi Yoshino (funded by Kakenhi, Japan)
3/2009–3/2011	Takehisa Hasegawa (funded by JST, Presto)

As host researcher (at least 1 month)

University of Michigan

3/1/2026–2/28/2027	Hang-Hyun Jo (Associate Professor on sabbatical, The Catholic University of Korea, South Korea)

State University of New York at Buffalo

6/5/2023–8/4/2023	Carlos Vázquez-de Jesus (undergraduate student, University of Puerto Rico at Cayey)
1/18/2023–2/25/2023 & 12/27/2024–2/17/2025	Jiyoung Kang (Assistant Professor, Pukyong National University, South Korea)
8/24/2021–4/21/2022 & 9/28/2022–3/21/2023	Kazuki Nakajima (PhD student, Tokyo Institute of Technology, Japan)

University of Bristol, United Kingdom

5/13/2019–8/31/2019	Bruna Campos Guedes (undergraduate student, Polytech Marseille — Aix-Marseille Université, France)
9/13/2018–3/12/2019	Shun Kodate (PhD student, Tohoku University, Japan) visit funded by Graduate Program in Data Science, Tohoku University
3/5/2018–4/6/2018	Jorge P. Rodríguez (PhD student, University of Balearic Islands, Spain) visit funded by Short Term Scientific Mission (STSM), COST Action: CA15109, EU Framework Programme Horizon 2020 on European Cooperation for Statistics of Network Data Science
1/19/2017–3/17/2017	Elohim Fonseca dos Reis (PhD student, University of Campinas, Brazil) visit funded by EPSRC, UK

University of Tokyo, Japan

7/16/2013–8/30/2013	Yanru Chen (undergraduate student, Tsinghua University, China) visit funded by Tsinghua University
6/11/2013–8/21/2013	Friederike Greb (postdoc, Georg-August-Universităt Gŏttingen, Germany) visit funded by Summer Program, Japan Society for the Promotion of Science
2/3/2012–3/2/2012	Paul Expert (postdoc, Imperial College, UK) visit funded by Postdoctoral Fellowships for Foreign Researchers (short-term), Japan Society for the Promotion of Science
5/1/2009–2/28/2010	Toshihiro Tanizawa (Associate Professor, Kochi National College of Technology, Japan) visit funded by KOSEN National Institute of Technology, Japan

PhD Thesis Examiner (except for students under my supervision)

2/2026–	Internal, PhD dissertation committee member, Zhiyuan (Stan) Yu Bioinformatics PhD Program, University of Michigan, USA
2/2026–	Internal, PhD dissertation committee member, Tiffany Wan Bioinformatics PhD Program, University of Michigan, USA
12/2025–	Internal, PhD committee member, Kaito Iwasaki Applied & Interdisciplinary Mathematics PhD Program, University of Michigan, USA
1/2026	Internal PhD examiner, Cameron Purple, Mathematical PhD Program, State University of New York at Buffalo, USA
4/2025	Internal PhD examiner, Tong Wu, Computational and Data-Enabled Sciences PhD Program, State University of New York at Buffalo, USA
11/2023–2/2026	External dissertation committee member, Xiliang Mao, Department of Civil and Environmental Engineering, University of South Florida, USA
4/2024	Pre-examiner, Javier Urena-Carrion, Department of Computer Science, Aalto University, Finland
8/2023	External PhD examiner, Kiefer Joe Burgess, Department of Management Sciences, University of Waterloo, Canada
5/2023	Internal PhD examiner, Bengier Ülgen Kılıç, Mathematics PhD Program, State University of New York at Buffalo, USA
11/2021	Internal PhD examiner, Zhao Song, Mathematics PhD Program, State University of New York at Buffalo, USA
5/2021	External PhD evaluator, Pitambar Khanra, Department of Mathematics, National Institute of Technology Durgapur, India
4/2021	External PhD examiner, Xingru Chen, Department of Mathematics, Dartmouth College, USA
5/2019	External PhD examiner, Prateek Verma, Department of Physical Sciences, Indian Institute of Science Education and Research (IISER) Kolkata, India
1/2019	External PhD examiner, Jeroen van Lidth de Jeude, PhD Program in Economics, Management and Data Science, IMT School of Advanced Studies, Italy also serving as member (secretary) of the Graduation Committee for Jeroen van Lidth de Jeude, Federica Parisi, and Giacomo Rapisardi, 03/2019
9/2018	External PhD examiner (president), Jorge P. Rodríguez, Doctoral Program in Physics, University of Balearic Islands, Spain
5/2018	External PhD examiner, Neil Sherborne, Department of Mathematics, University of Sussex, UK
12/2017	External PhD examiner, Se-Wook Oh, University of Oxford, UK
10/2017	External PhD examiner, Balasundaram Kadirvelu, School of Biological Sciences, Biomedical Engineering Section, University of Reading, UK
7/2016	Internal PhD examiner, Christopher John McWilliams, Department of Engineering Mathematics, University of Bristol, UK
4/2016	Internal PhD examiner, Holly Silk, Department of Engineering Mathematics, University of Bristol, UK
2008–2014	PhD examiner for 20 candidates University of Tokyo, Department of Mathematical Informatics NB: By convention, no notion of internal/external examiner
2011–2014	PhD examiner for 6 candidates, University of Tokyo (other departments)

Miscellaneous

Languages: Japanese (native), English (fluent), and Spanish (fluent).

Herd member: Saving Endangered Species Int'l Playwriting Prize (winning plays announced 02/2016)

Last updated: August 18, 2026